

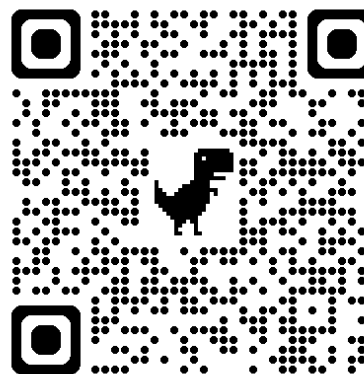


自然語言處理與應用

Natural Language Processing and Applications

BERT and its Family

Instructor: 林英嘉 (Ying-Jia Lin)
2026/03/30



GitHub



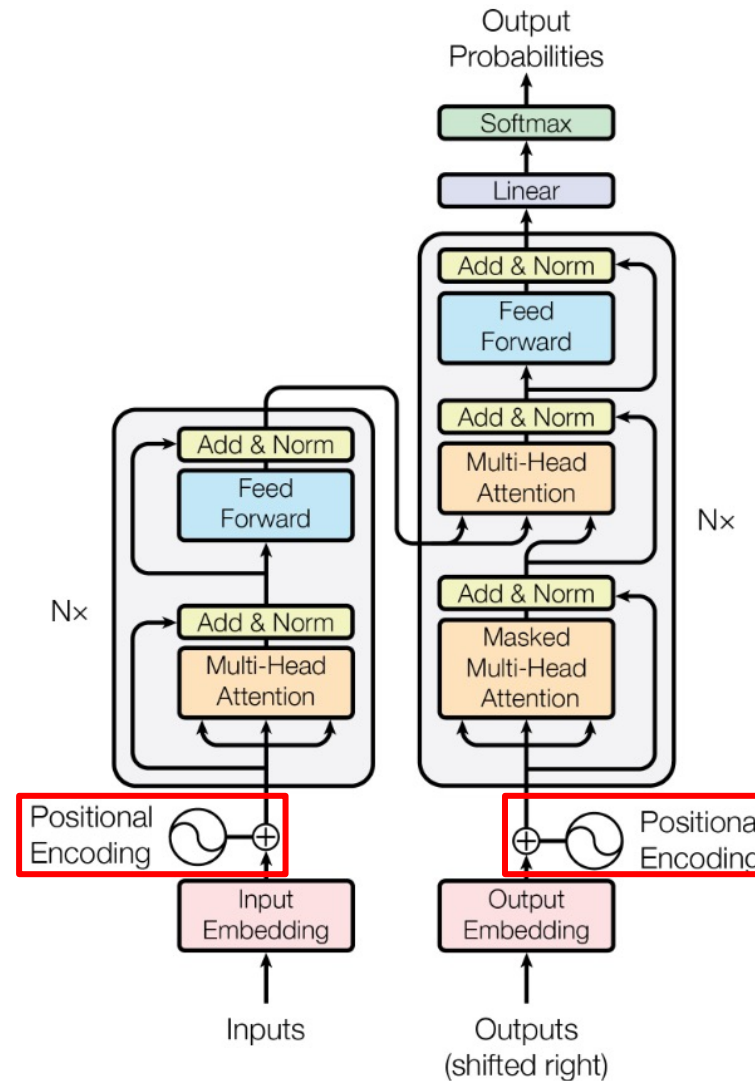
Slido
NLP0330
([Link](#))

Outline

- Recap [15 min]
- Positional Encoding [20 min]
- Cross-attention [15 min]
- BERT and its Family [50 min]
- Code [35 min]
- Quiz [15 min]

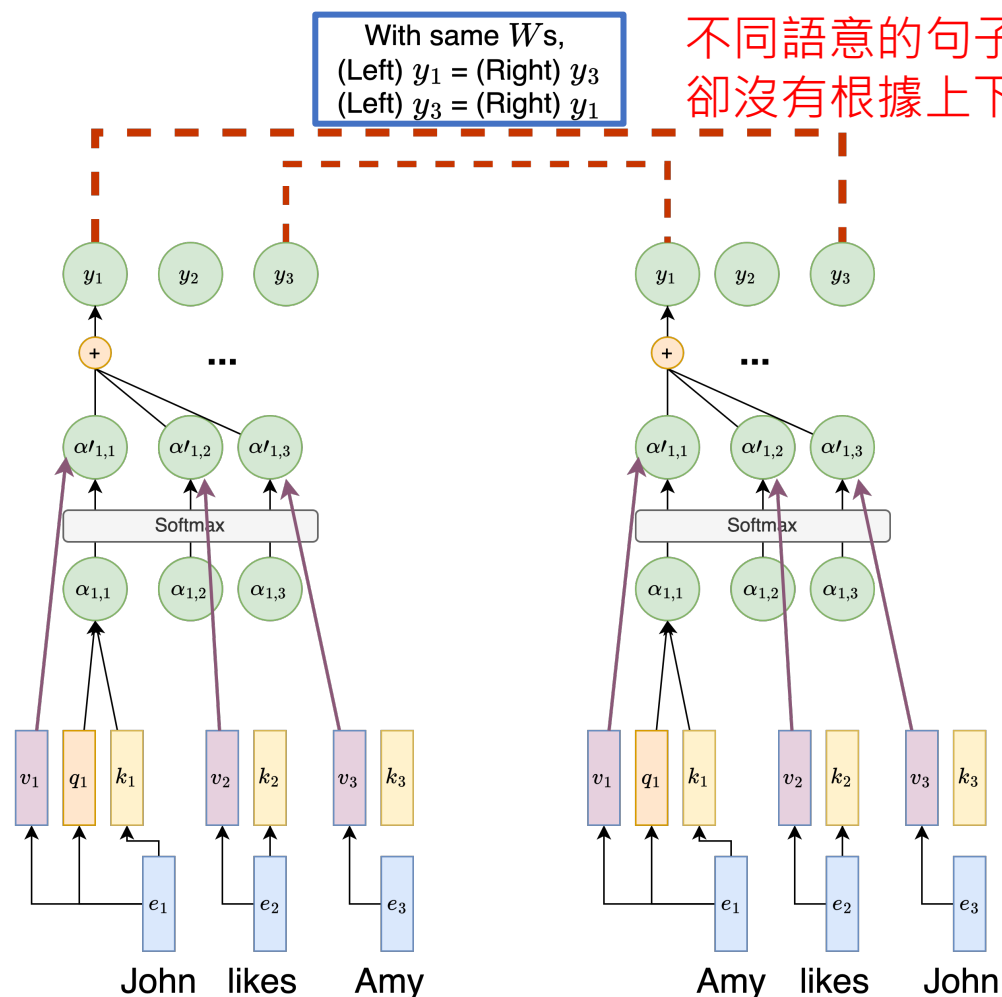
Outline of Transformer

Vaswani, Ashish, et al. "Attention is all you need." NeurIPS (2017).

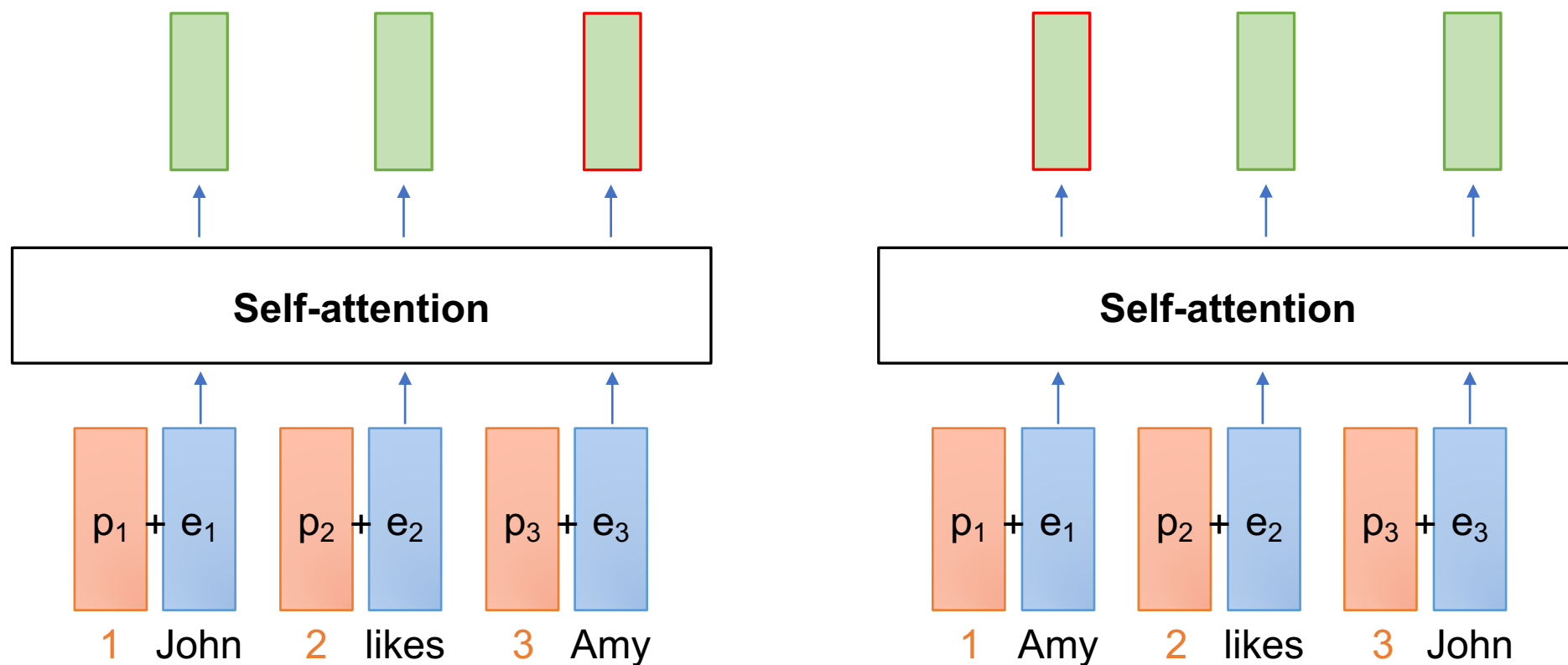


Lack of Position Modeling

- RNN 有位置資訊 (因為 RNN step-by-step 處理文字)
- 然而，Transformer 中 self-attention 的操作遺失了 step-by-step 處理文字的方式，故缺少位置資訊



常見位置資訊作法：Learned Positional Embeddings (PE)



此做法又稱絕對位置資訊編碼 (Absolute positional Encoding)

Learned Positional Embeddings 程式碼

https://huggingface.co/transformers/v2.1.1/_modules/transformers/modeling_bert.html

```
class BertEmbeddings(nn.Module):
    """Construct the embeddings from word, position and token_type embeddings.
    """
    def __init__(self, config):
        super(BertEmbeddings, self).__init__()
        self.word_embeddings = nn.Embedding(config.vocab_size, config.hidden_size, padding_idx=0)
        self.position_embeddings = nn.Embedding(config.max_position_embeddings, config.hidden_size)
        self.token_type_embeddings = nn.Embedding(config.type_vocab_size, config.hidden_size)

        # self.LayerNorm is not snake-cased to stick with TensorFlow model variable name and be able to load
        # any TensorFlow checkpoint file
        self.LayerNorm = BertLayerNorm(config.hidden_size, eps=config.layer_norm_eps)
        self.dropout = nn.Dropout(config.hidden_dropout_prob)
```

缺點：Learned Positional Embeddings 不會超出 max_position 之後的資訊

BERT PE 範例

假設 `max_position_embeddings = 5`，`hidden_size = 4`，那麼 `nn.Embedding(5, 4)` 會建立一張這樣的表：

位置 0 → [0.12, -0.34, 0.56, 0.78]

位置 1 → [0.91, 0.23, -0.45, 0.67]

位置 2 → [-0.11, 0.88, 0.33, -0.52]

位置 3 → [0.44, -0.76, 0.21, 0.95]

位置 4 → [-0.63, 0.15, 0.82, -0.37]

當輸入句子 "我 喜歡 NLP" 時，三個 token 分別在位置 0、1、2：

"我" → 查位置 0 → [0.12, -0.34, 0.56, 0.78]

"喜歡" → 查位置 1 → [0.91, 0.23, -0.45, 0.67]

"NLP" → 查位置 2 → [-0.11, 0.88, 0.33, -0.52]

Positional Encoding in Transformer

- 以絕對位置編碼創造出相對位置的 embeddings
- 核心精神：利用三角函數的週期性來建構序列資訊

$$PE_{(pos, 2i)} = \sin(\underline{pos} / 10000^{2i/d_{\text{model}}})$$
$$PE_{(pos, 2i+1)} = \cos(\underline{pos} / 10000^{2i/d_{\text{model}}})$$

i 代表 embedding 的位置， d_{model} 是 embedding 長度
10000 是實驗得出的較佳數值

輸入序列中任一 token 的位置

Transformer PE 範例

分母：
 $i = 0 \rightarrow 100^0 = 1$
 $i = 1 \rightarrow 100^{1/2} = 10$

$$PE_{(pos, 2i)} = \sin\left(\frac{pos \cdot 100^{2i/d_{model}}}{d_{model}}\right) \quad \text{設 } d_{model} = 4$$
$$PE_{(pos, 2i+1)} = \cos\left(\frac{pos \cdot 100^{2i/d_{model}}}{d_{model}}\right)$$

I
love
nlp
.

		$i = 0$		$i = 1$	
	token	$\sin(t/1)$	$\cos(t/1)$	$\sin(t/10)$	$\cos(t/10)$
t=0	I	0.00	1.00	0.00	1.00
t=1	love	0.84	0.54	0.10	1.00
t=2	nlp	0.91	-0.42	0.20	0.98
t=3	.	0.14	-0.99	0.30	0.96

I 的 PE

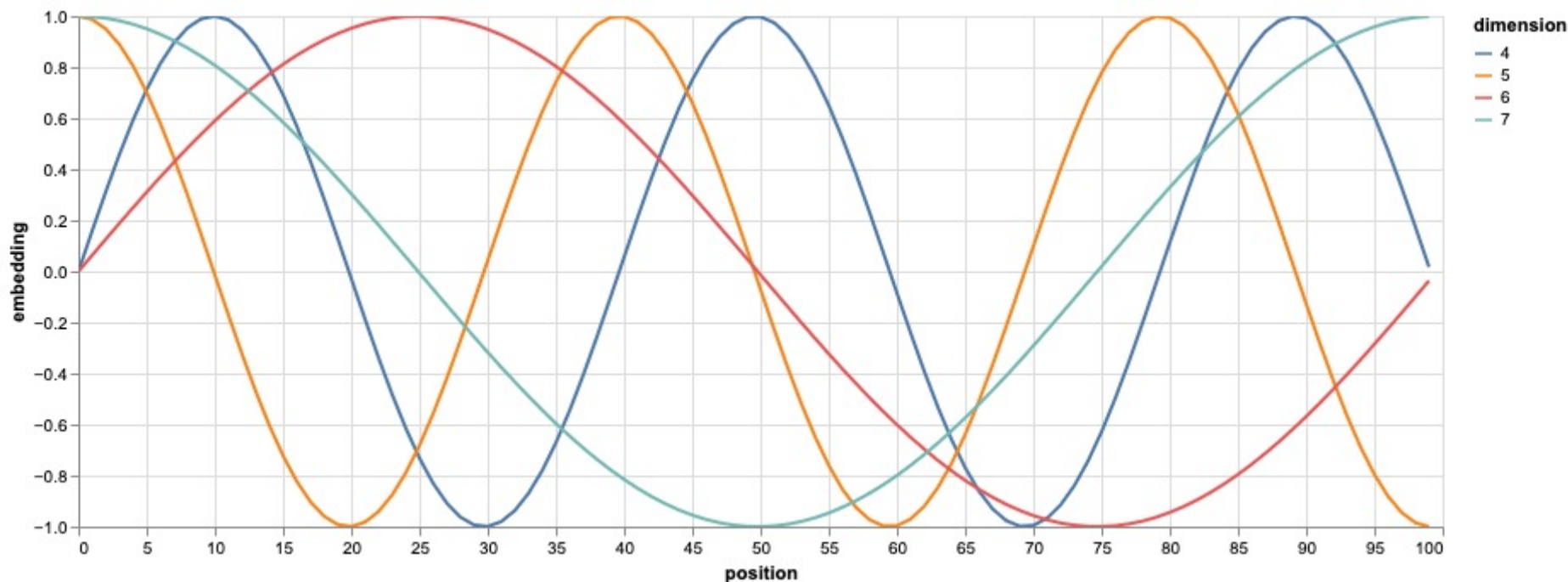
love 的 PE

nlp 的 PE

. 的 PE

Positional Encoding Visualization

i 越大 -> 分母越大 -> 相位變化越慢 -> 頻率越小



- Question: 為什麼需要同時有 Sin 和 Cosine 的函數？
- Ans: 藉由不同週期性來捕捉序列中各個位置的關係

$$\sin 30^\circ = 0.5$$

$$\sin 150^\circ = 0.5$$

vs.

$$(\sin 30^\circ, \cos 30^\circ) = (0.5, 0.866)$$

$$(\sin 150^\circ, \cos 150^\circ) = (0.5, -0.866)$$

為什麼需要同時有 Sin 和 Cos 函數？

設 $d_{\text{model}} = 2$

$$\text{PE}(t) = \begin{bmatrix} \sin(\omega t) \\ \cos(\omega t) \end{bmatrix}, \omega = 1/10000^{2i/d_{\text{model}}} \quad (\text{為求公式簡潔，本頁會省略 } \omega)$$

$$\text{PE}(t + k) = \begin{bmatrix} \cos(k) & \sin(k) \\ -\sin(k) & \cos(k) \end{bmatrix} \begin{bmatrix} \sin(t) \\ \cos(t) \end{bmatrix} = M(k) \cdot \text{PE}(t)$$

旋轉矩陣

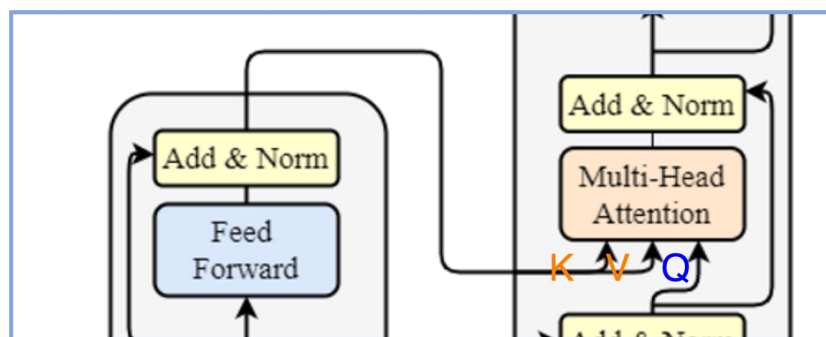
$$\text{score} = (\text{t時間點的input})W_Q \cdot W_K^T (\text{t+k時間點的input})^T$$

$$= (x_t + \underline{\text{PE}(t)})W_Q \cdot W_K^T (x_{t+k} + \underline{\text{PE}(t+k)})^T$$

$$\Rightarrow \text{PE}(t) \cdot W_Q \cdot W_K^T \cdot \text{PE}(t+k)^T = \underline{\text{PE}(t)} \cdot W_Q \cdot W_K^T \cdot (\underline{M(k) \cdot \text{PE}(t)})^T$$

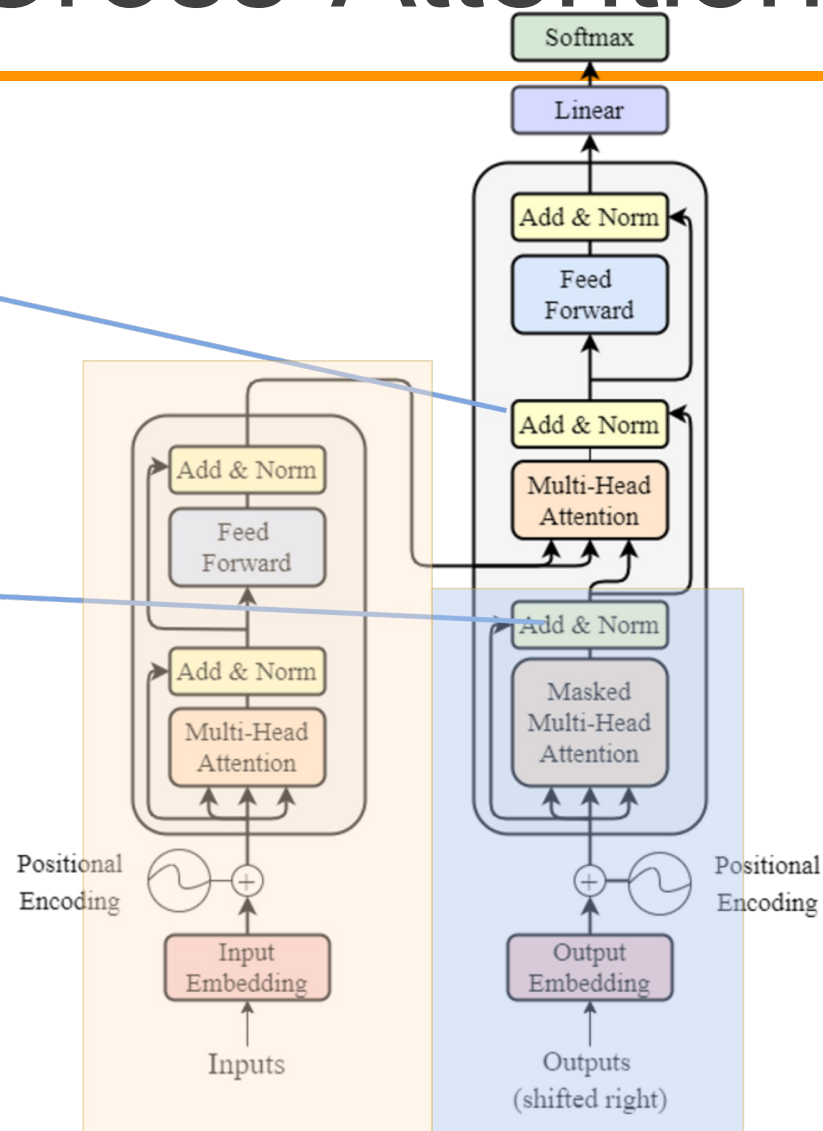
💡 讓模型在計算 attention 時能留意相對位置 (角度) 關係

Encoder-Decoder 橋樑: Cross-Attention



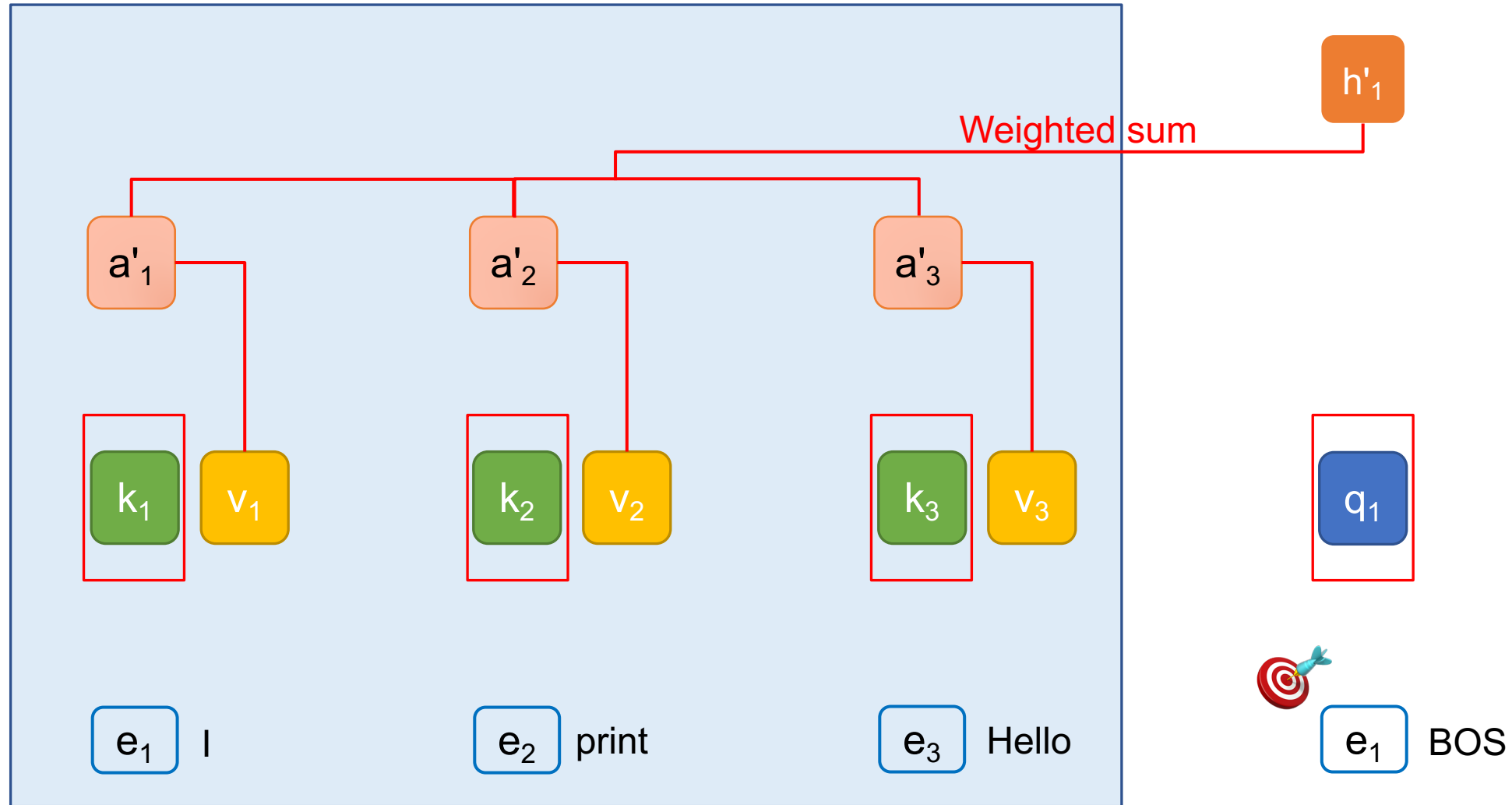
K, V uses the output from encoder as their X;
Q uses the decoder's information as X.

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$



Cross-attention

Encoder



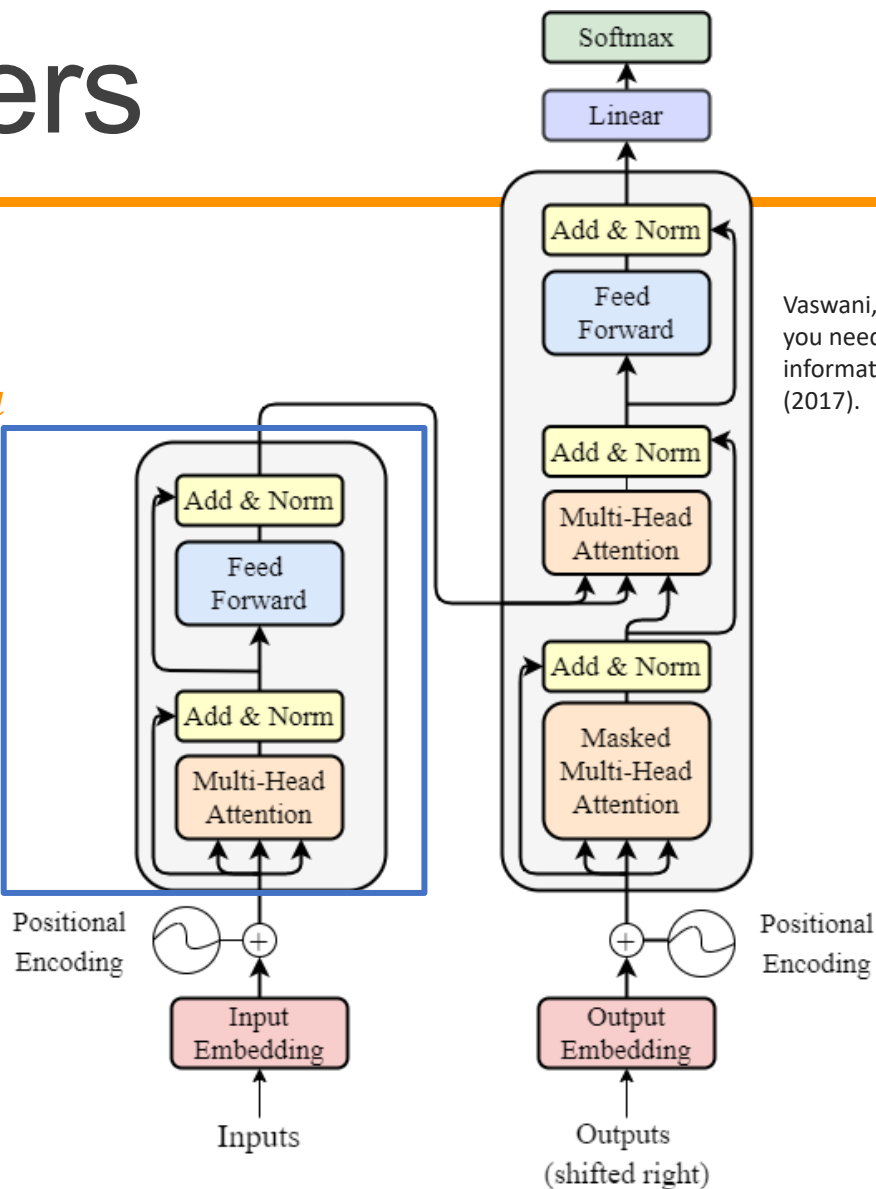
Number of Layers

Input, output dimension match!
We can have multiple encoders stacked up, to increase the number of trainable parameters.

$$\text{output} \in \mathbb{R}^{N \times d}$$

N *

$$\text{input} \in \mathbb{R}^{N \times d}$$



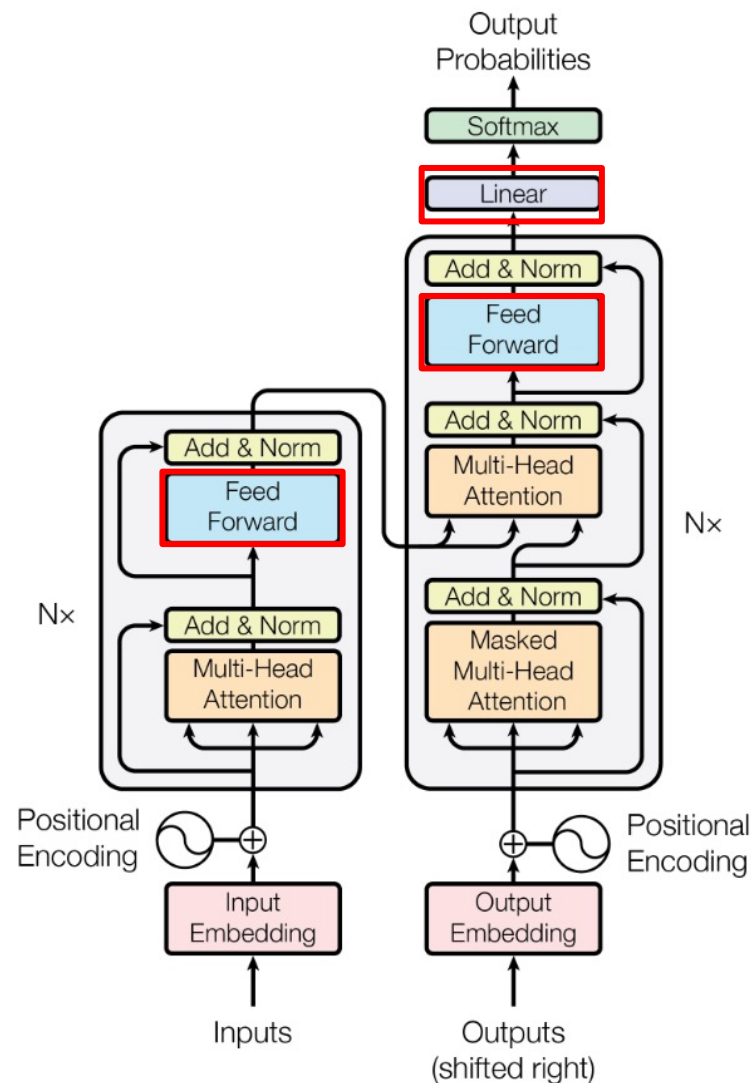
Vaswani, Ashish, et al. "Attention is all you need." Advances in neural information processing systems 30 (2017).

Outline of Transformer

Feed Forward Networks:

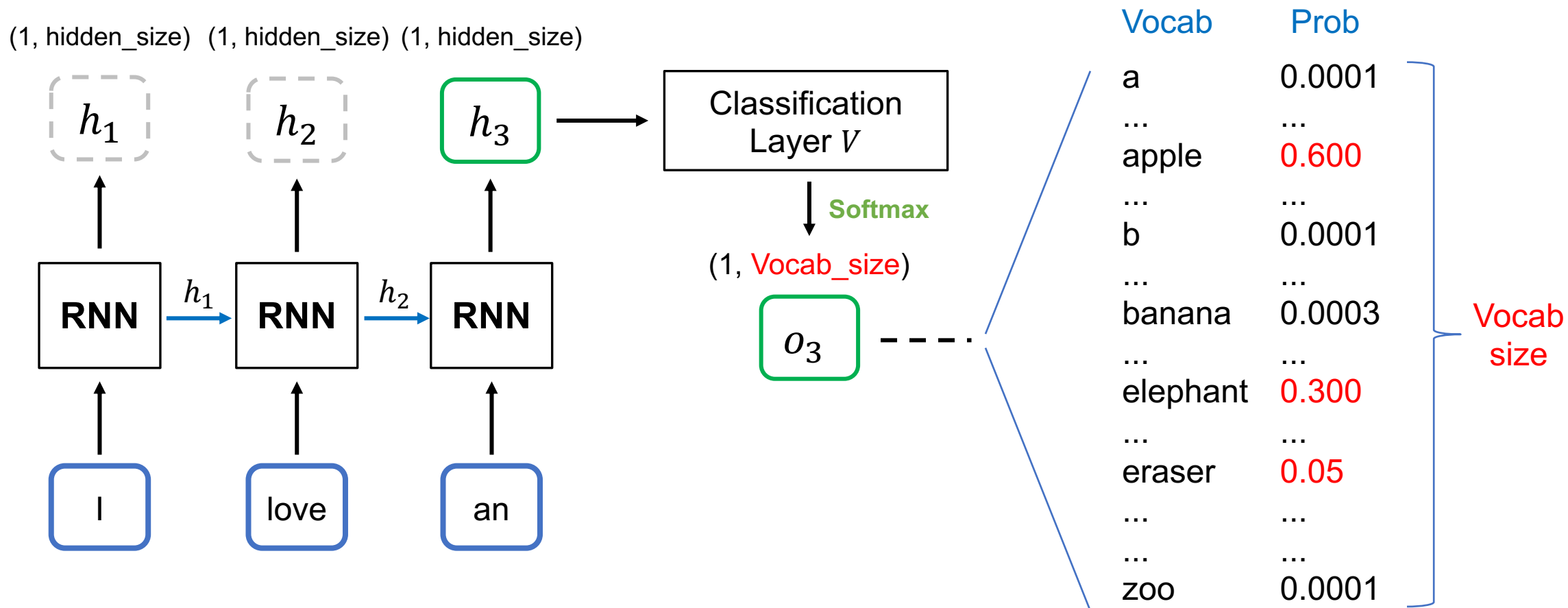
- Simply multiply by a weight matrix (MLP).
- Used to project to a specific dimension.

Vaswani, Ashish, et al. "Attention is all you need." NeurIPS (2017).



[Recap] Text Generation with RNN

*本頁的RNN不包含輸出層 V



Transformers' Achievements

1. In original paper, State-of-The-Art (SoTA) of machine translation.

Table 2: The Transformer achieves better BLEU scores than previous state-of-the-art models on the English-to-German and English-to-French newstest2014 tests at a fraction of the training cost.

Model	BLEU		Training Cost (FLOPs)	
	EN-DE	EN-FR	EN-DE	EN-FR
ByteNet [18]	23.75			
Deep-Att + PosUnk [39]		39.2		$1.0 \cdot 10^{20}$
GNMT + RL [38]	24.6	39.92	$2.3 \cdot 10^{19}$	$1.4 \cdot 10^{20}$
ConvS2S [9]	25.16	40.46	$9.6 \cdot 10^{18}$	$1.5 \cdot 10^{20}$
MoE [32]	26.03	40.56	$2.0 \cdot 10^{19}$	$1.2 \cdot 10^{20}$
Deep-Att + PosUnk Ensemble [39]		40.4		$8.0 \cdot 10^{20}$
GNMT + RL Ensemble [38]	26.30	41.16	$1.8 \cdot 10^{20}$	$1.1 \cdot 10^{21}$
ConvS2S Ensemble [9]	26.36	41.29	$7.7 \cdot 10^{19}$	$1.2 \cdot 10^{21}$
Transformer (base model)	27.3	38.1	$3.3 \cdot 10^{18}$	
Transformer (big)	28.4	41.8	$2.3 \cdot 10^{19}$	

Vaswani, Ashish, et al. "Attention is all you need." Advances in neural information processing systems 30 (2017).

6 encoders,
6 decoders,
h = 512, 100K train
steps.

6 encoders,
6 decoders,
h = 1024, 300K
train steps.

Transformer Variants (there are many more)

- Universal Transformer

- Adds Adaptive Computation Time
- Dehghani, Mostafa, et al. "Universal transformers." arXiv preprint arXiv:1807.03819 (2018).

- Longformer

- Tackles long documents
- Iz Beltagy, et al. (2020). Longformer: The Long-Document Transformer. arXiv:2004.05150.

- Roformer

- Changes the design of positional encoding
- Su, Jianlin, et al. "RoFormer: Enhanced Transformer with Rotary Position Embedding. CoRR abs/2104.09864 (2021)." arXiv preprint arXiv:2104.09864 (2021).

- Vision Transformer

- Tackles vision tasks
- Alexey Dosovitskiy, et al. "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale." International Conference on Learning Representations. 2021.

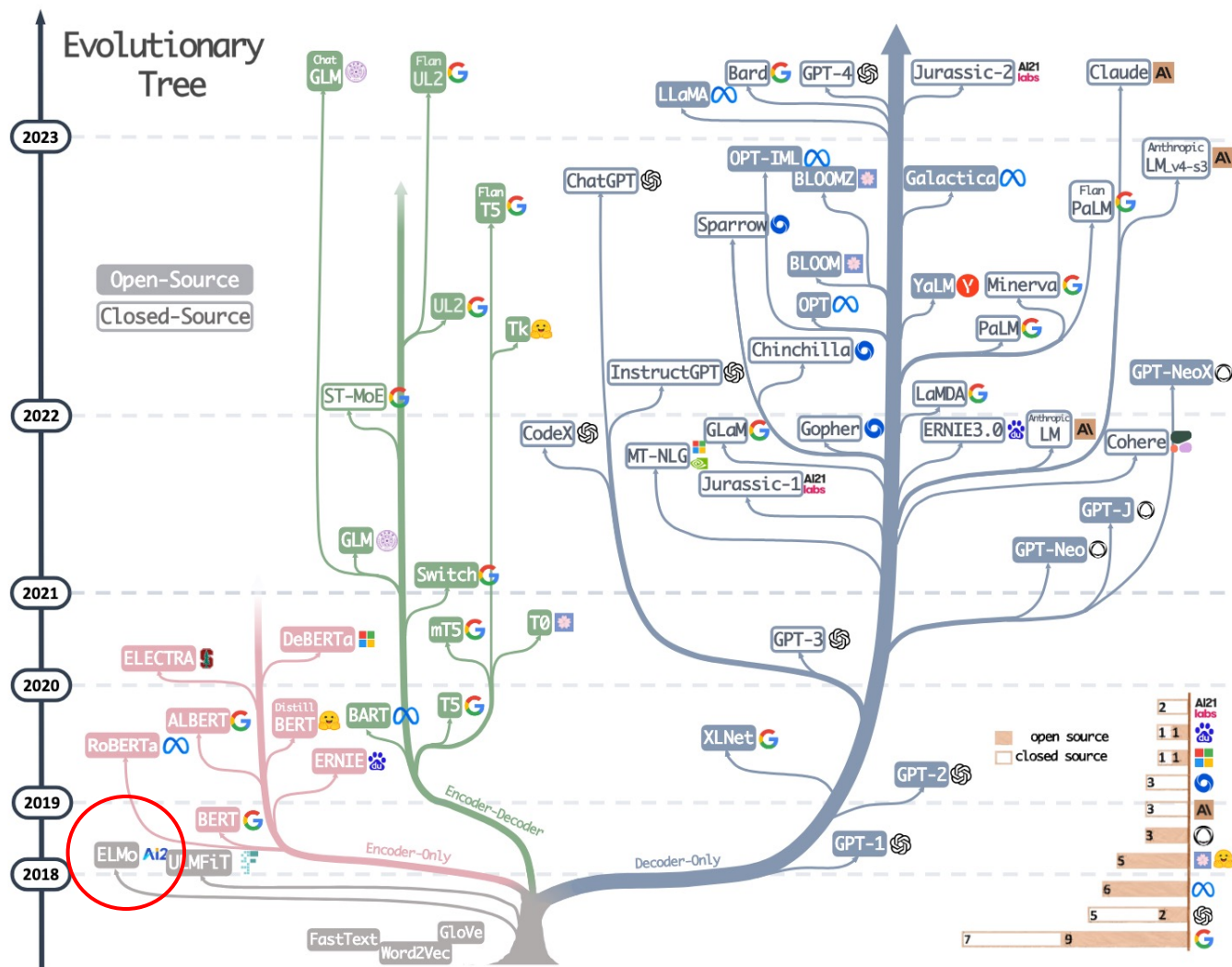
Additional Links

- <https://nlp.seas.harvard.edu/annotated-transformer/>
- <https://datascience.stackexchange.com/questions/82451/why-is-10000-used-as-the-denominator-in-positional-encodings-in-the-transformer>

BERT and its Family

- ELMo
- BERT
- RoBERTa
- SpanBERT
- DeBERTa

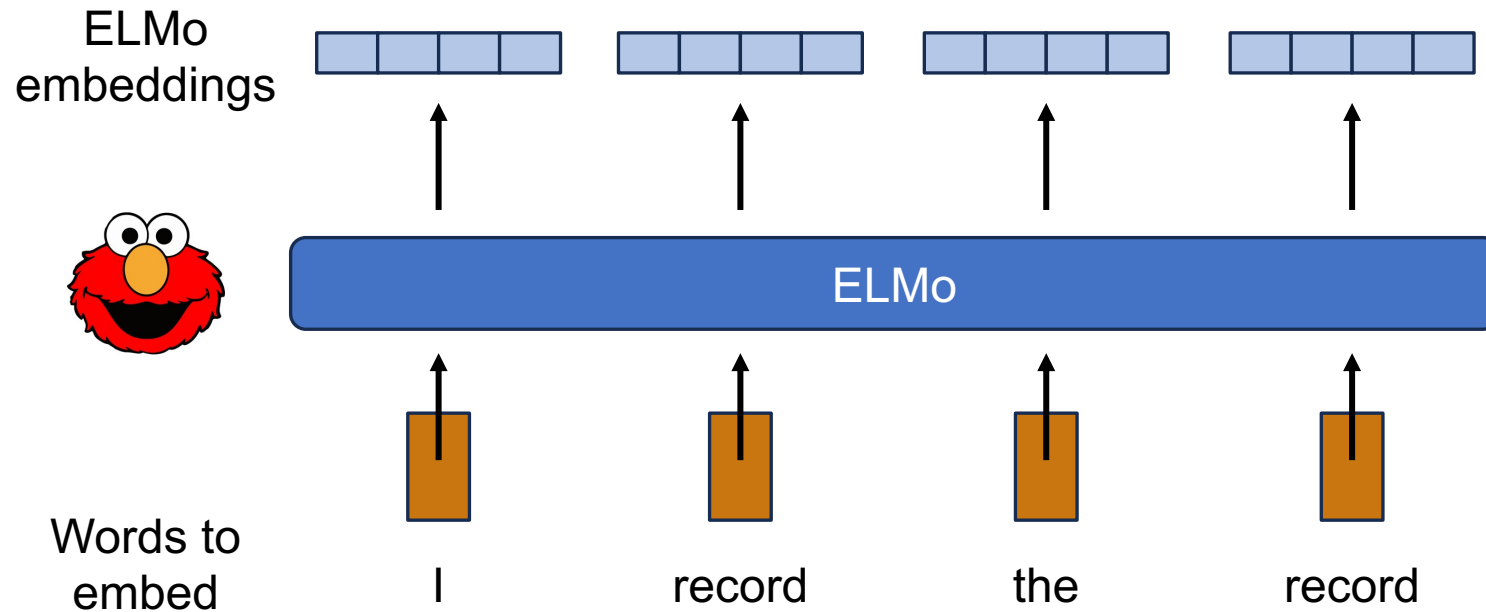
Evolutionary Tree of Large Language Models



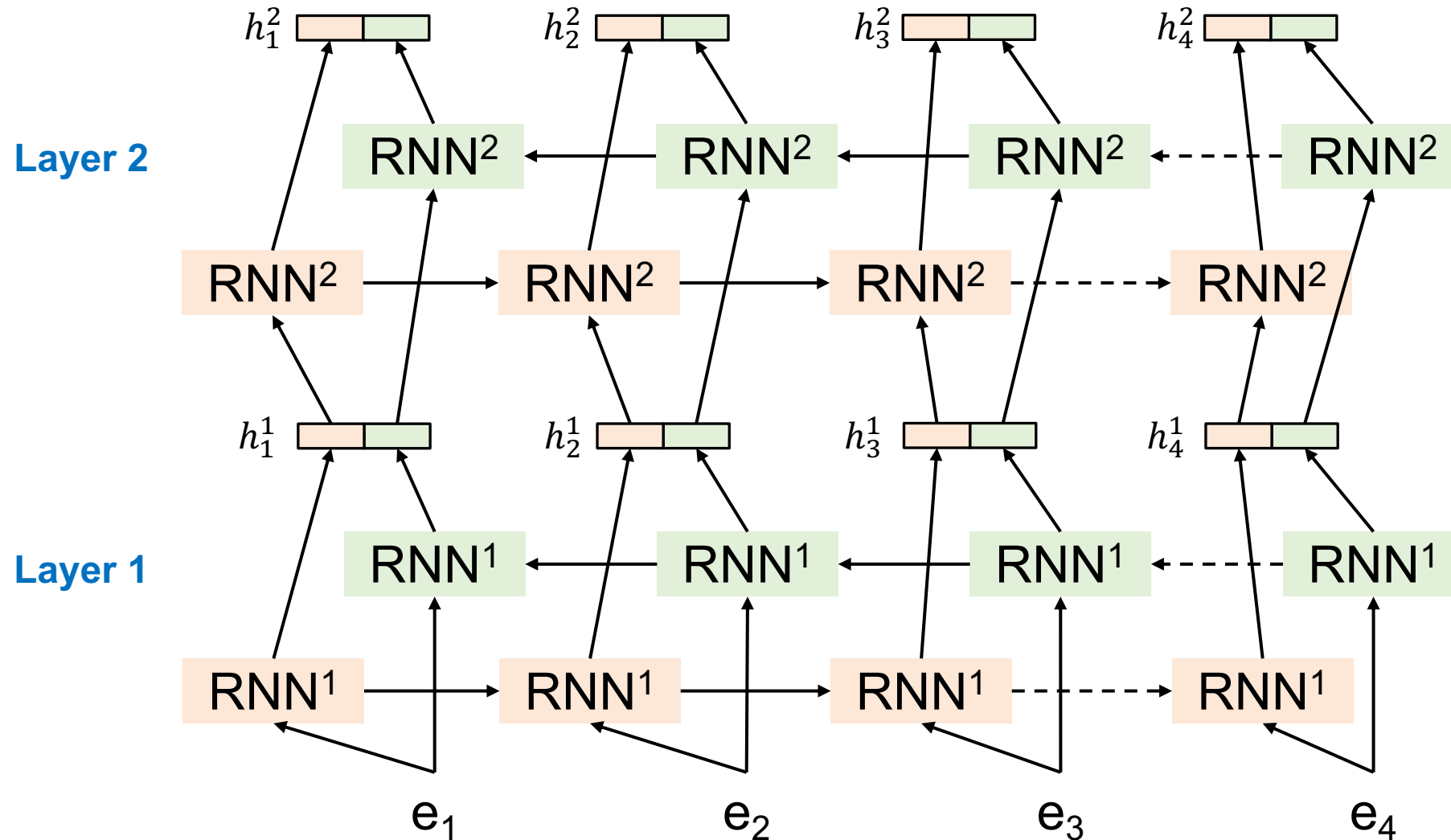
Yang, Jingfeng, et al. "Harnessing the power of llms in practice: A survey on chatgpt and beyond." ACM Transactions on Knowledge Discovery from Data 18.6 (2024): 1-32.

Context Matters: ELMo (Peters et al., 2018)

Rather than using a fixed representation for each word, ELMo considers the entire sentence before assigning each word in it an embedding.

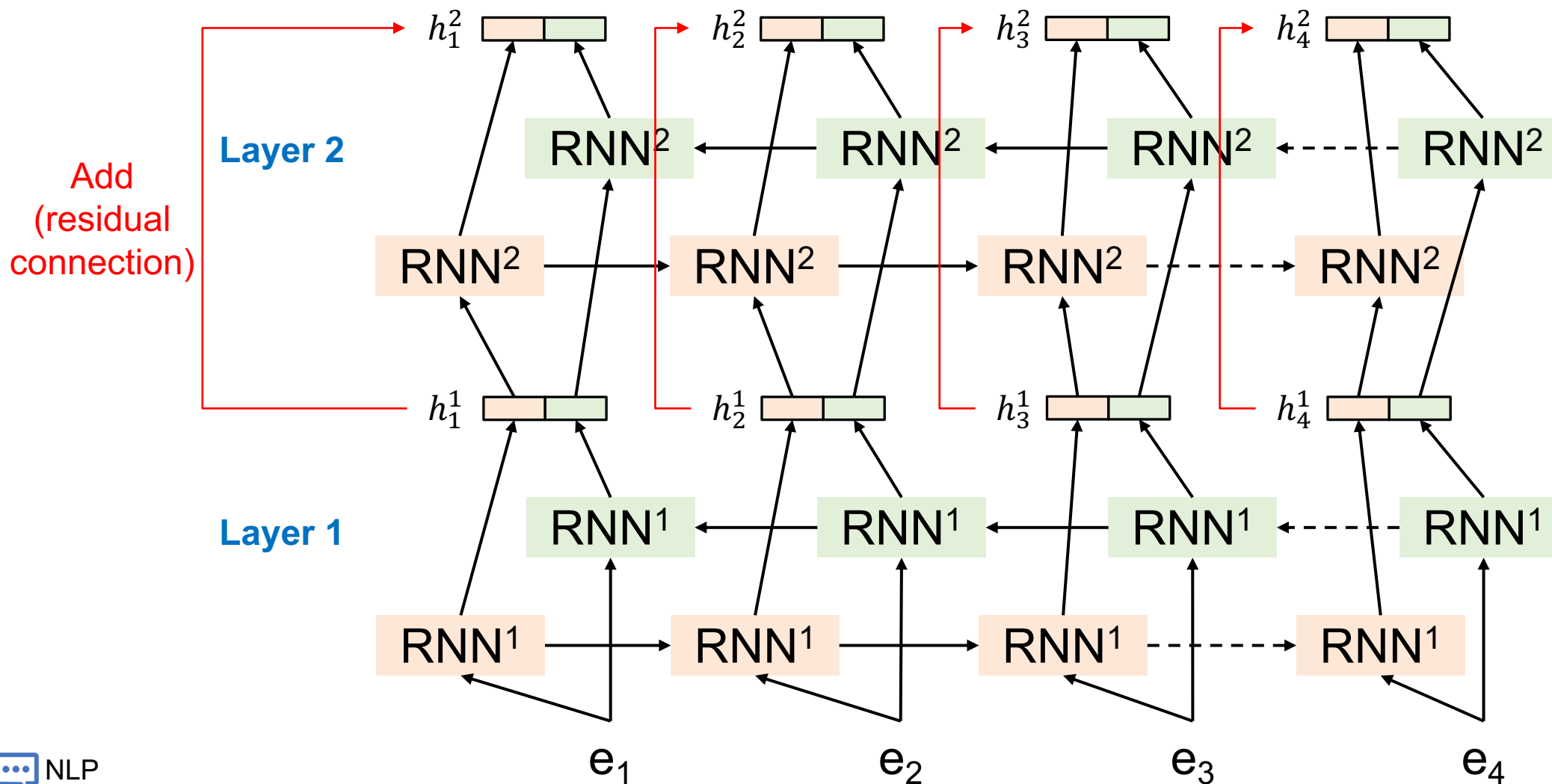


Bidirectional RNN with 2 Layers

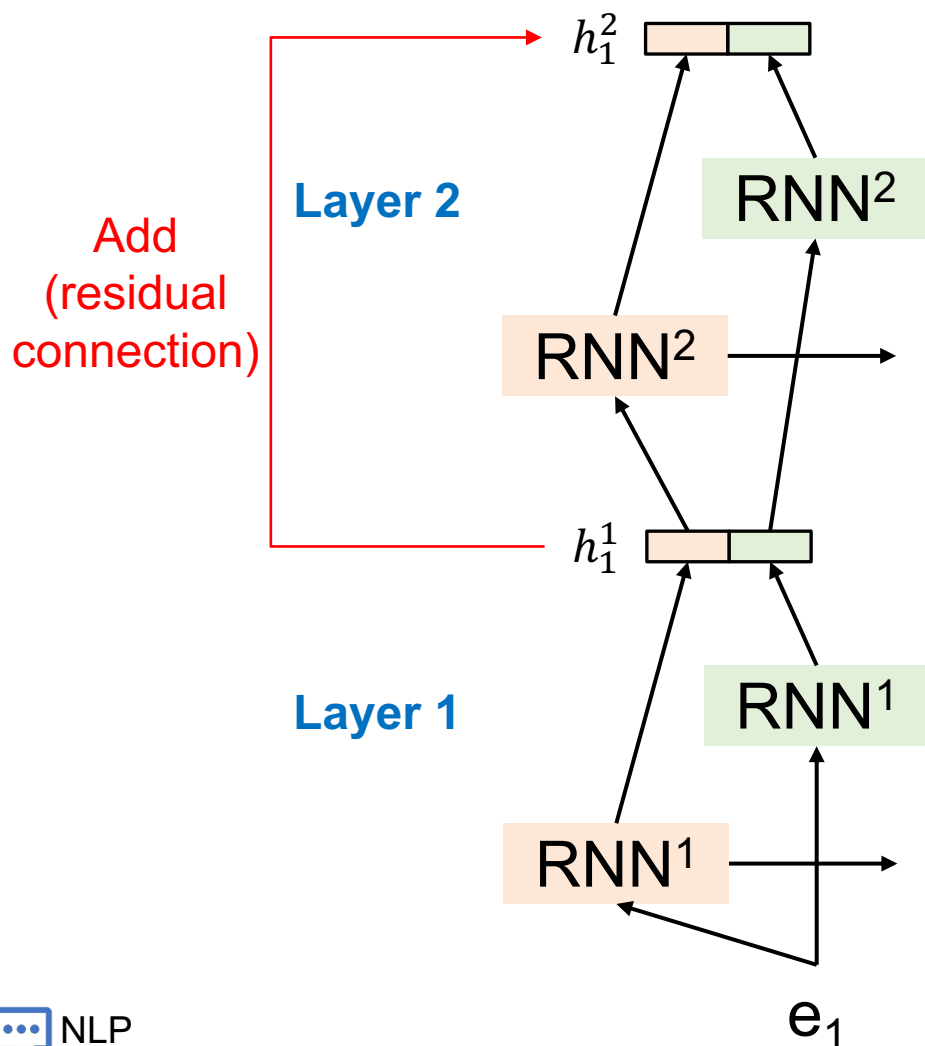


ELMo (1)

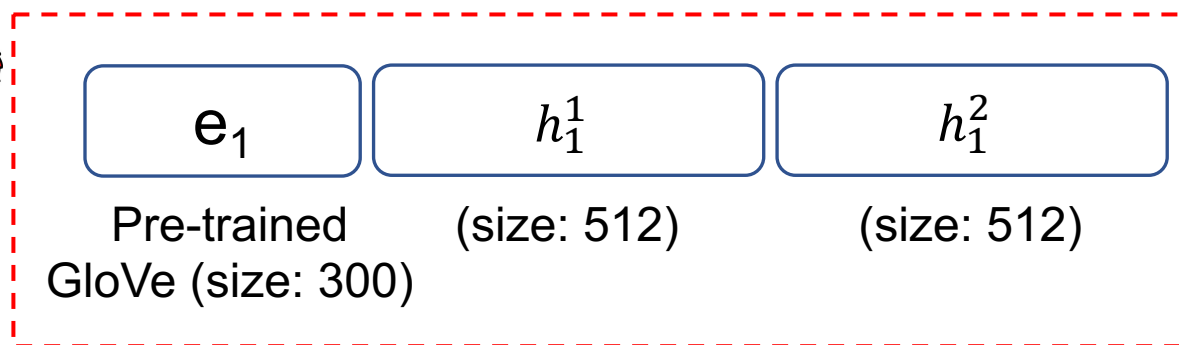
- ELMo 是一個兩層的Bi-LSTM
- 在第1層和第2層之間有: residual connections



ELMo (2)

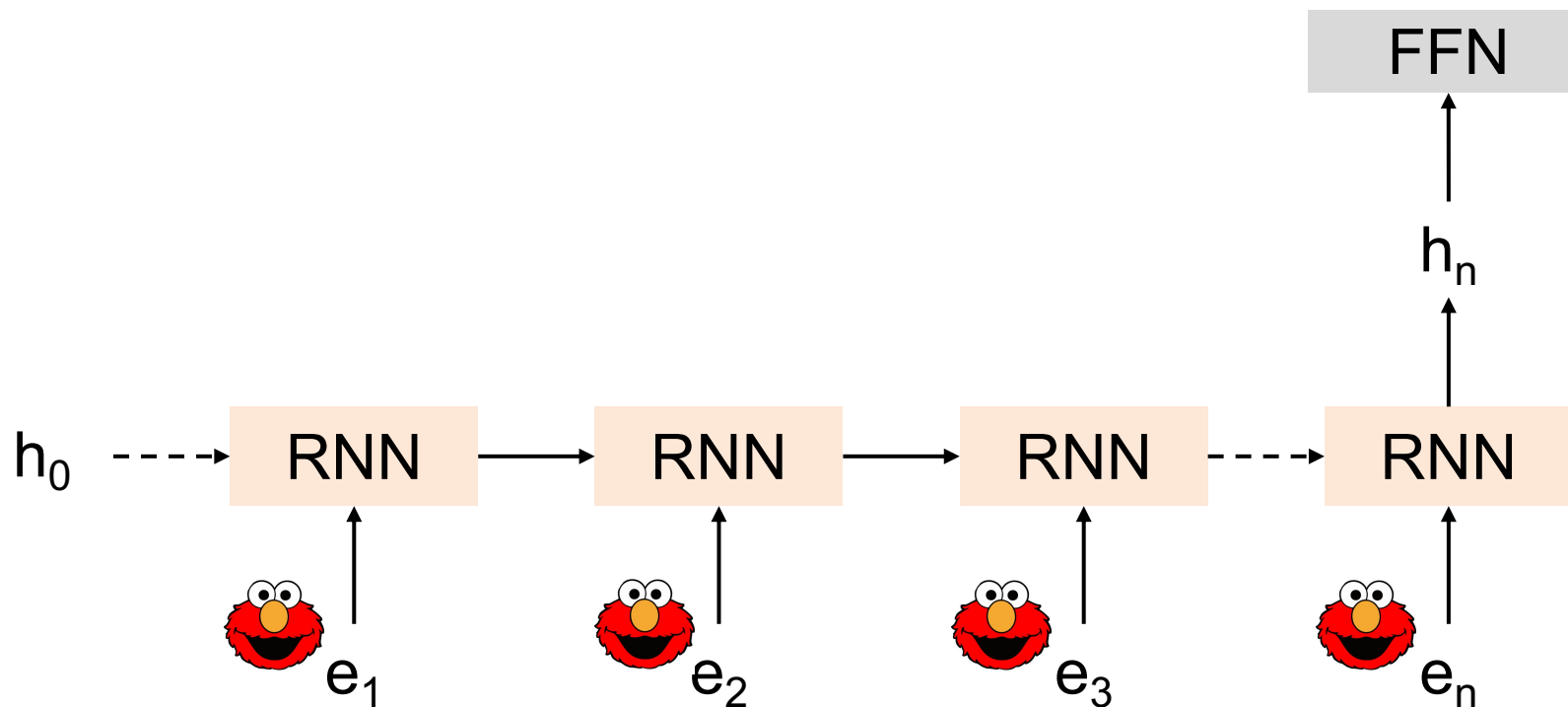
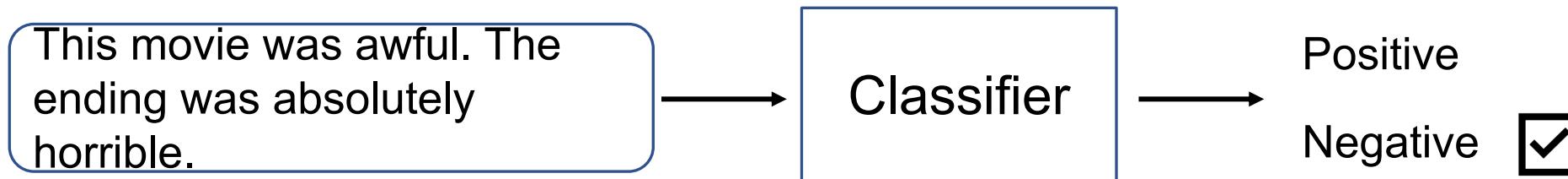


- ELMo 的輸出是 embedding
- ELMo 是 contextual embedding
 - 每個 token 的 embedding 由 input embedding 和每一層的 hidden state 做 concatenation



token₁ 的 token embedding (size: 1324)

ELMo 的使用方法



ELMo 相較於詞嵌入確實具備上下文語意資訊

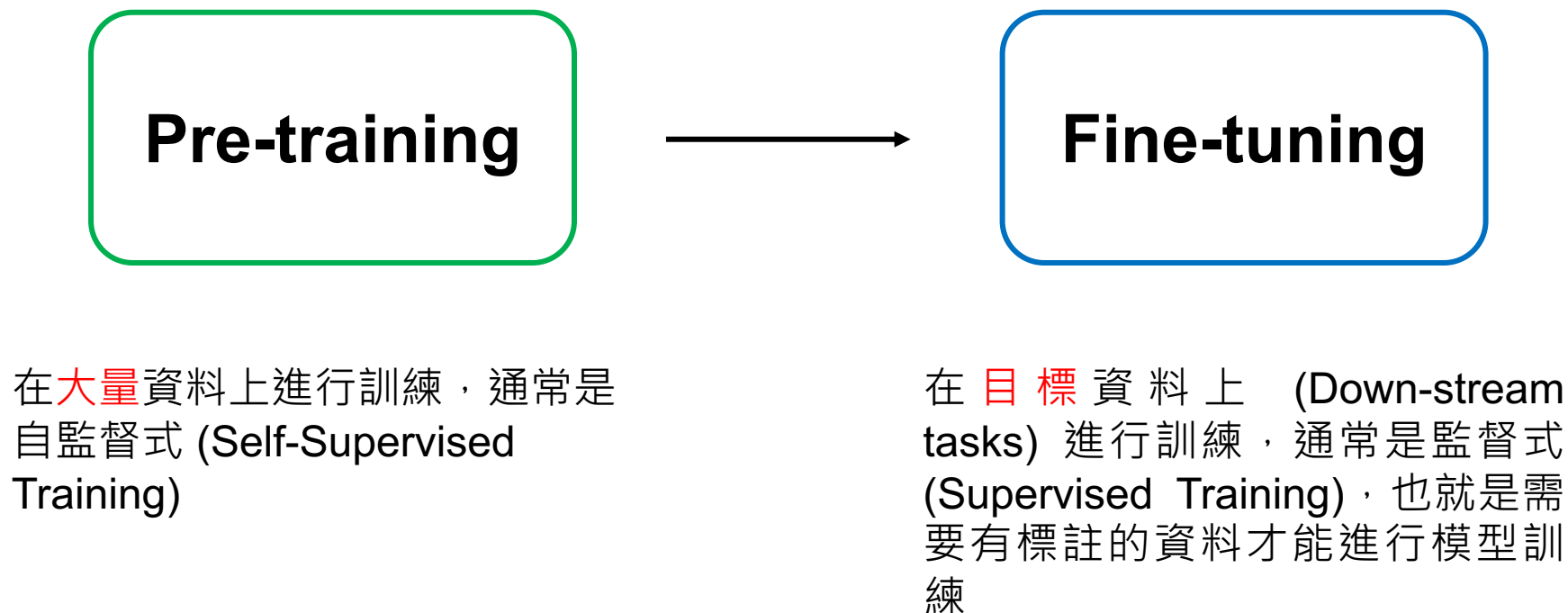
	Source	Nearest Neighbors
GloVe	play	playing, game, games, played, players, plays, player, Play, football, multiplayer
biLM	Chico Ruiz made a spectacular <u>play</u> on Alusik 's grounder {...}	Kieffer , the only junior in the group , was commended for his ability to hit in the clutch , as well as his all-round excellent <u>play</u> .
	Olivia De Havilland signed to do a Broadway <u>play</u> for Garson {...}	{...} they were actors who had been handed fat roles in a successful <u>play</u> , and had talent enough to fill the roles competently , with nice understatement .

跟球類有關
的 play

跟表演有關
的 play

Table 4: Nearest neighbors to “play” using GloVe and the context embeddings from a biLM.

The Pre-training then Fine-tuning Paradigm



Feature-based and Fine-tuning approaches

- Feature-based approach – ELMo
 - ELMo 並無針對下游任務設計模型結構
 - ELMo 是一種外掛、增幅器，接在別人的工作上
- Fine-tuning approach – GPT, BERT
 - 涵蓋下游任務設計

Pre-trained Model

**New
Layer (s)**

Static Embedding vs. Contextual Embedding

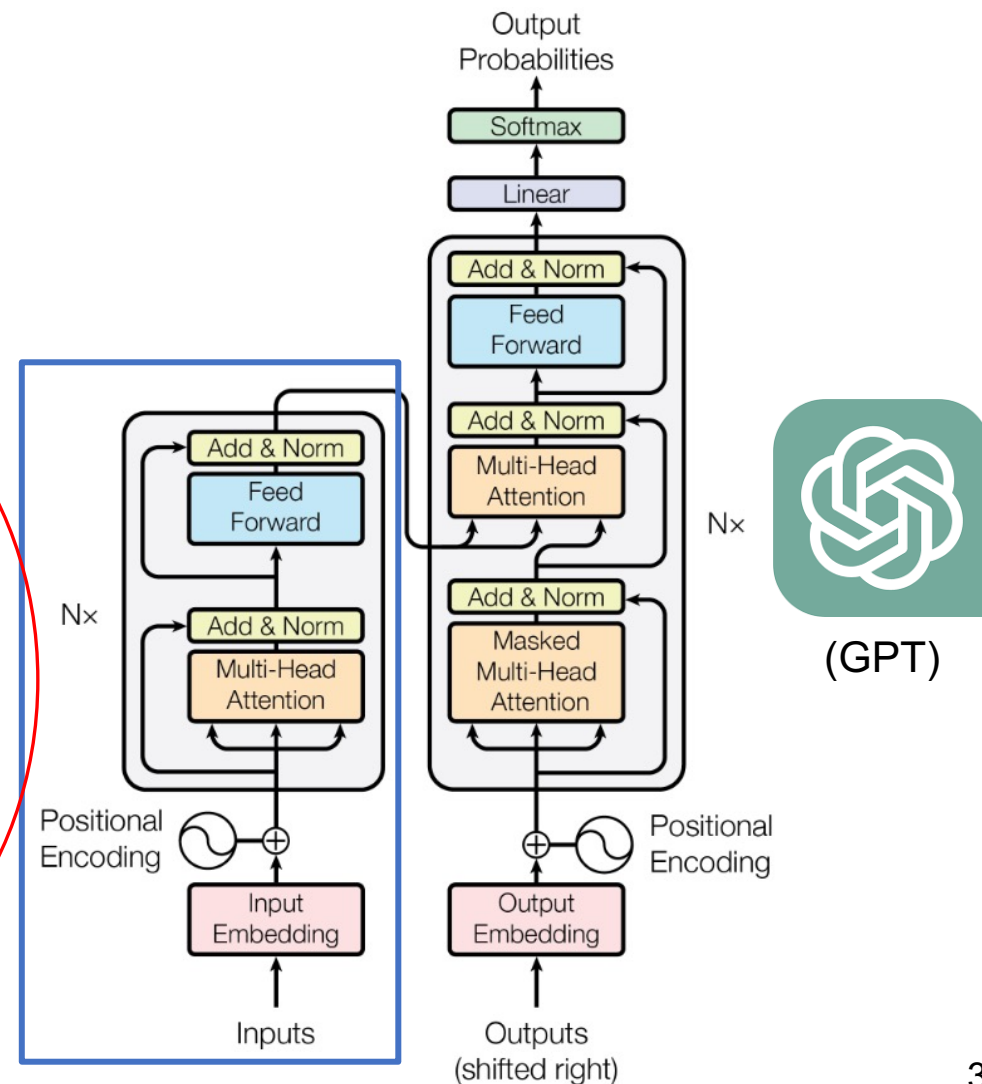
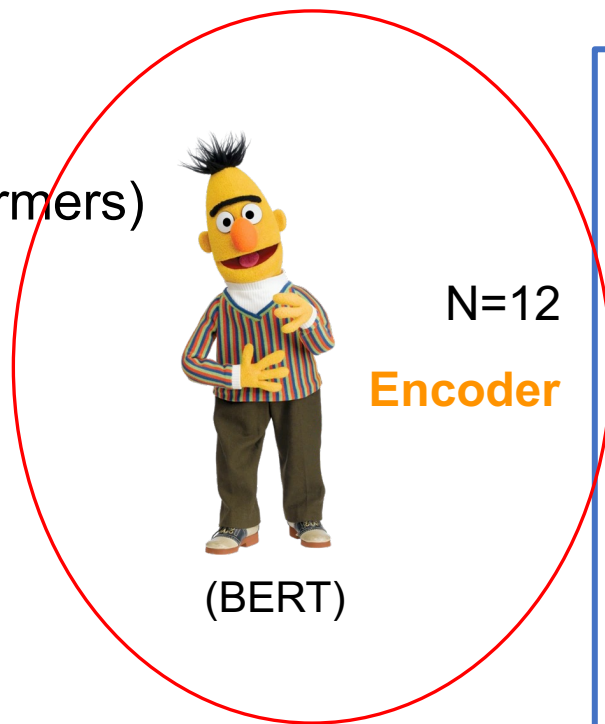
Type	Models
Static Embedding	Word2Vec, GloVe, ...
Contextual Embedding	RNN, ELMo, BERT, GPT, ...

BERT and its Family

- ELMo
- BERT
- RoBERTa
- SpanBERT
- DeBERTa

Transformer 架構的後續應用

- Generative Pre-training (GPT) series
- BERT (Bidirectional encoder representations from transformers)
 - [Devlin et al., 2018](#)



BERT 的輸入

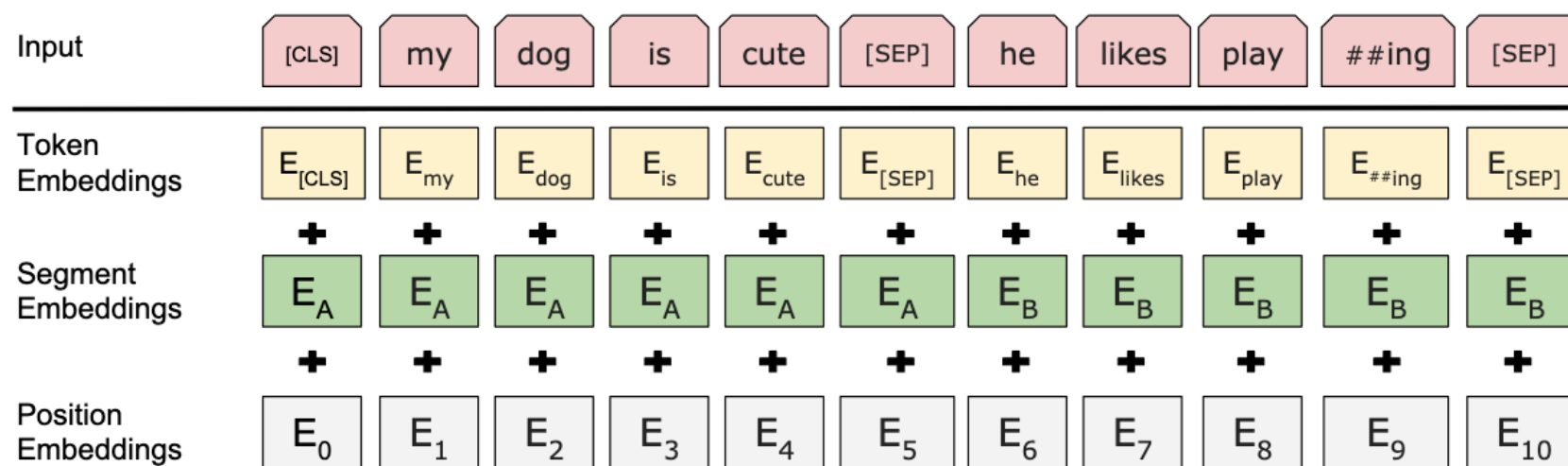
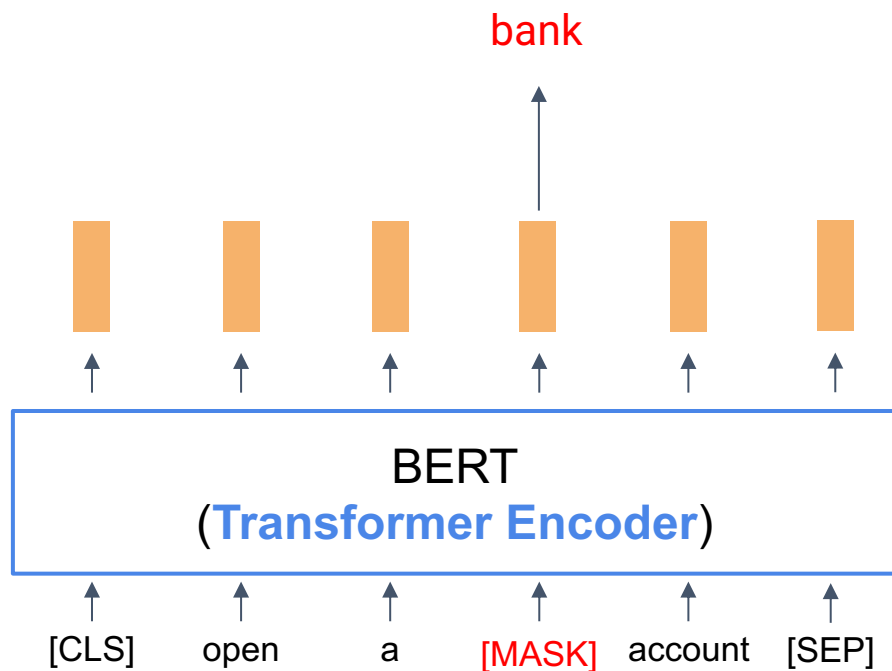


Figure 2: BERT input representation. The input embeddings are the sum of the token embeddings, the segmentation embeddings and the position embeddings.

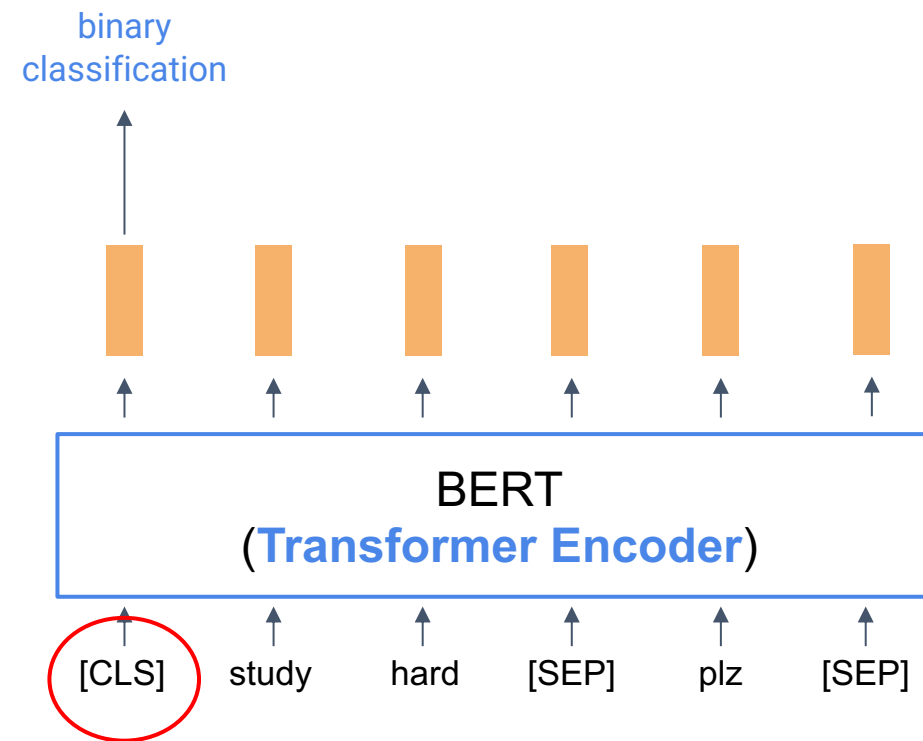
BERT (**B**idirectional Encoder Representations from Transformers)

- BERT 有兩種預訓練任務：

Masked Language Modelling



Next Sentence Prediction



Next Sentence Prediction (NSP)

- Binarized next sentence prediction task
- Aimed at making BERT better on understanding the relationship between two sentences

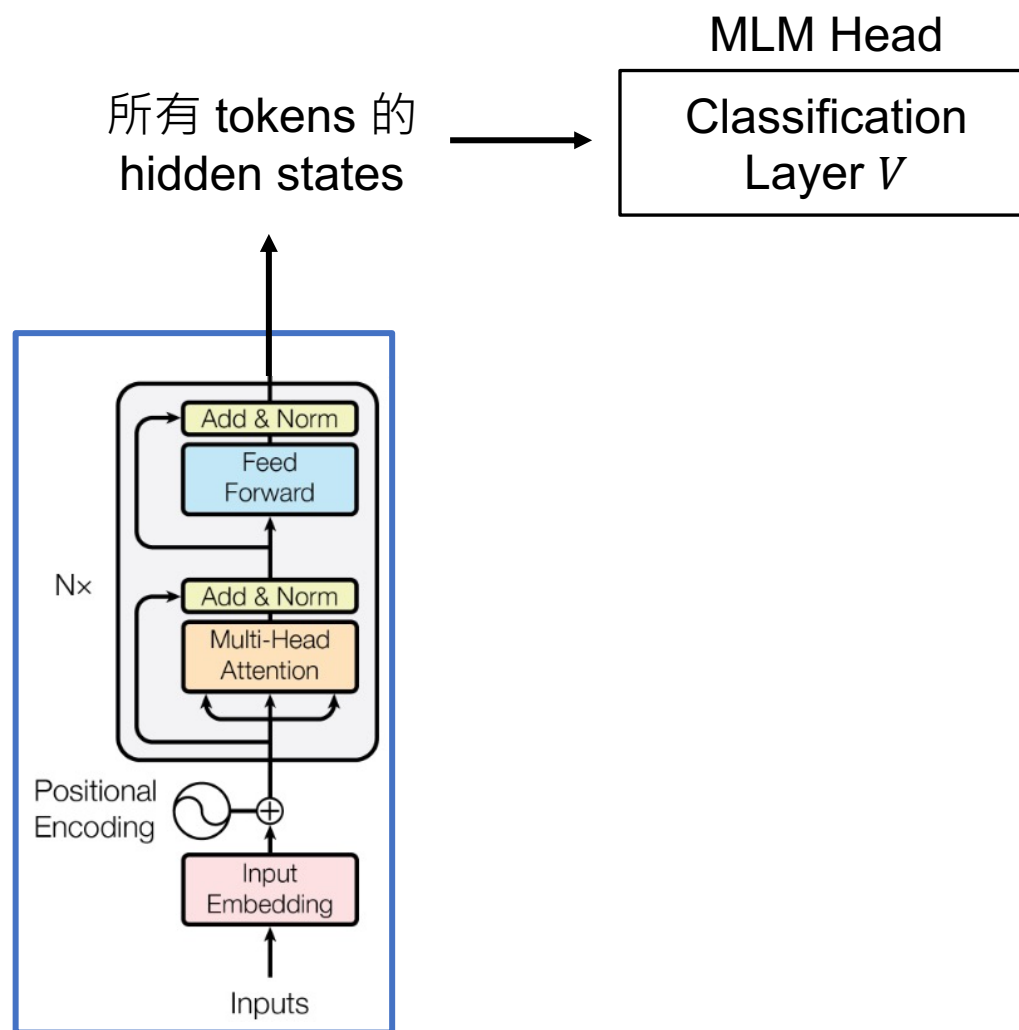
Input = [CLS] the man went to [MASK] store [SEP]
he bought a gallon [MASK] milk [SEP]

Label = `IsNext`

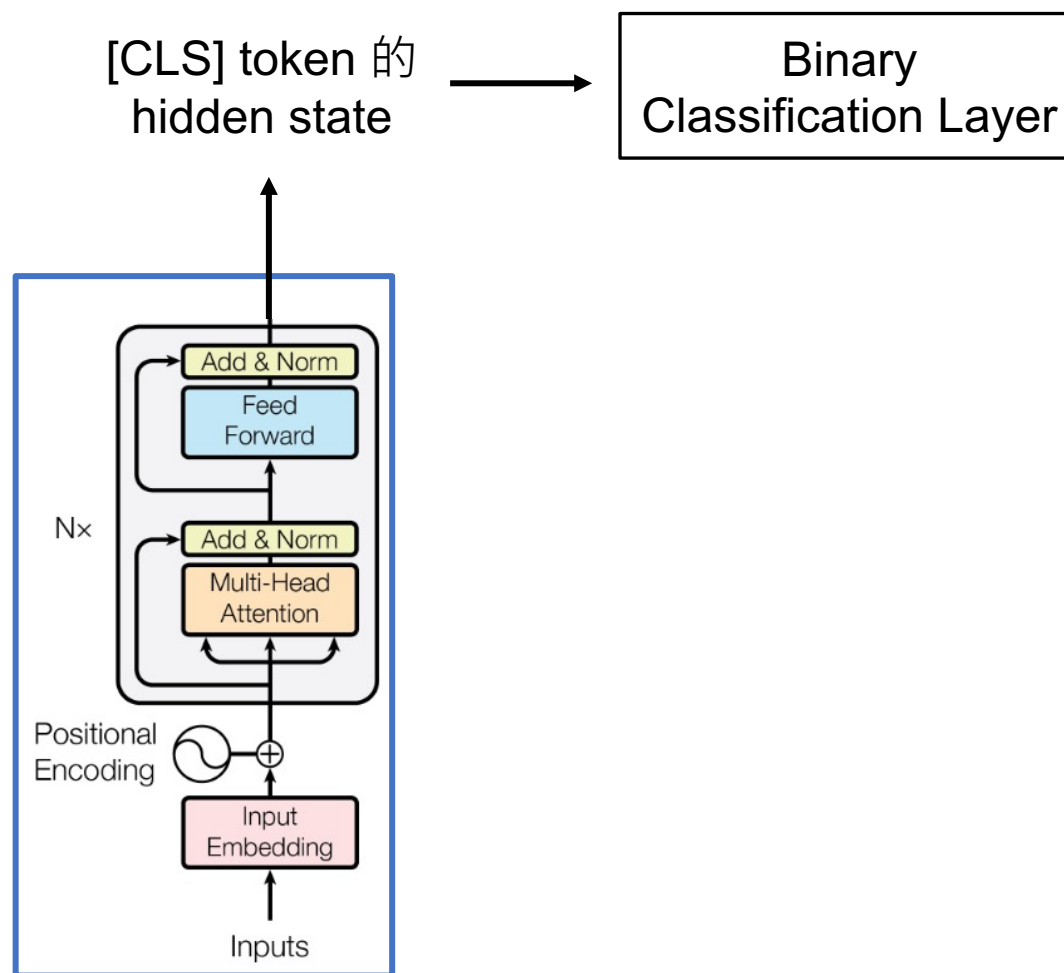
Input = [CLS] the man [MASK] to the store [SEP]
penguin [MASK] are flight ##less birds [SEP]

Label = `NotNext`

進行 MLM 時的 BERT 結構

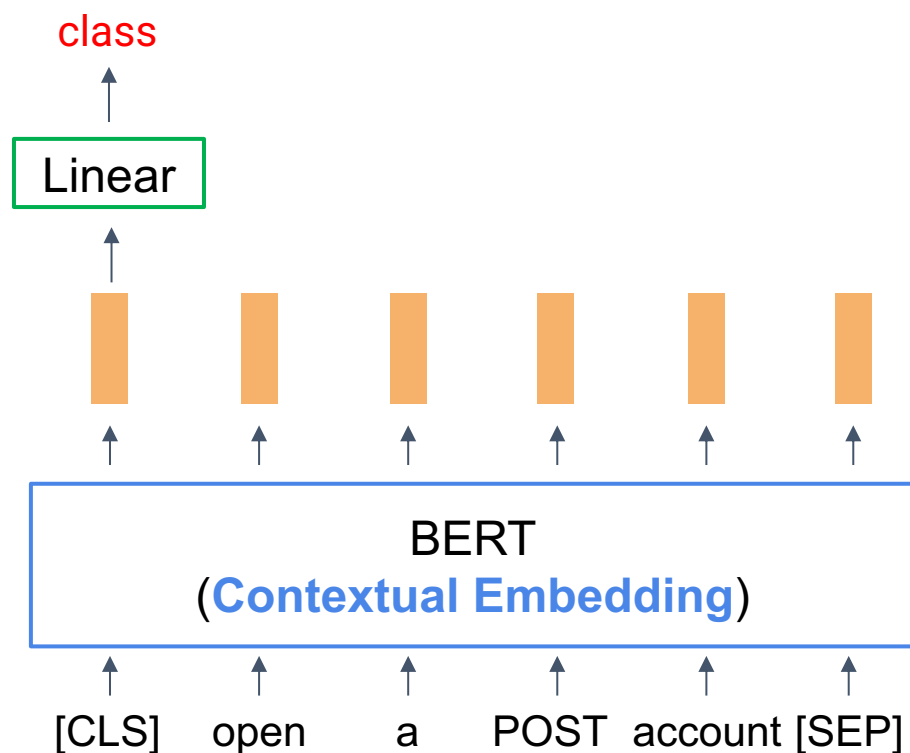


進行 NSP 時的 BERT 結構

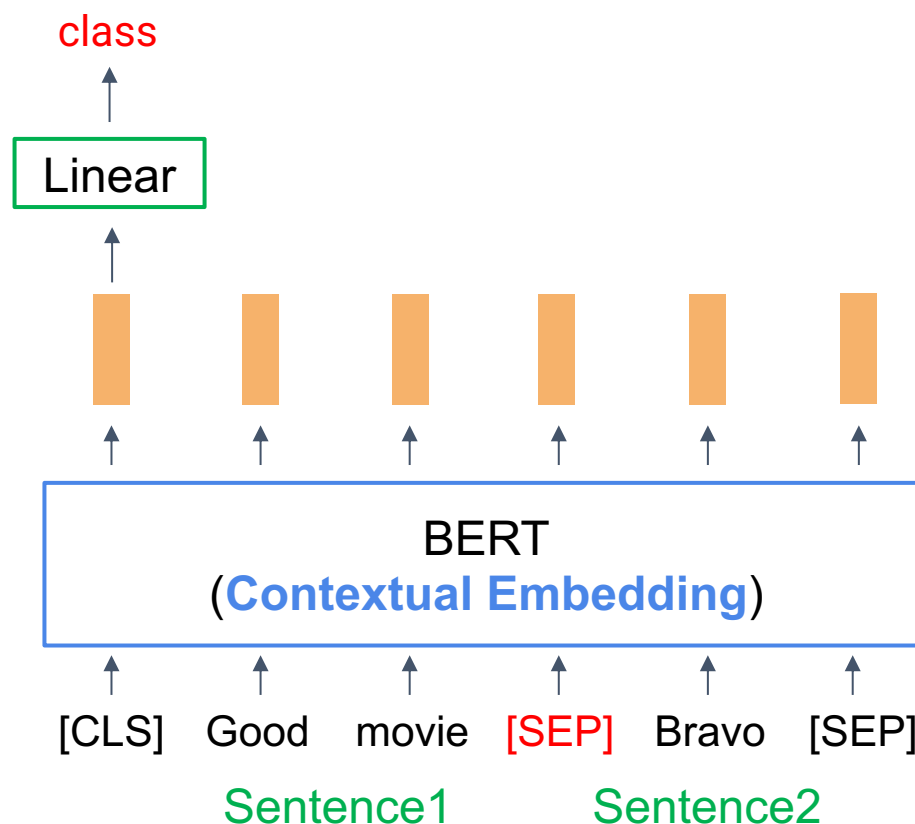


Fine-tuning BERT (Sentence Classification)

Single-sentence classification



Sentence pair classification



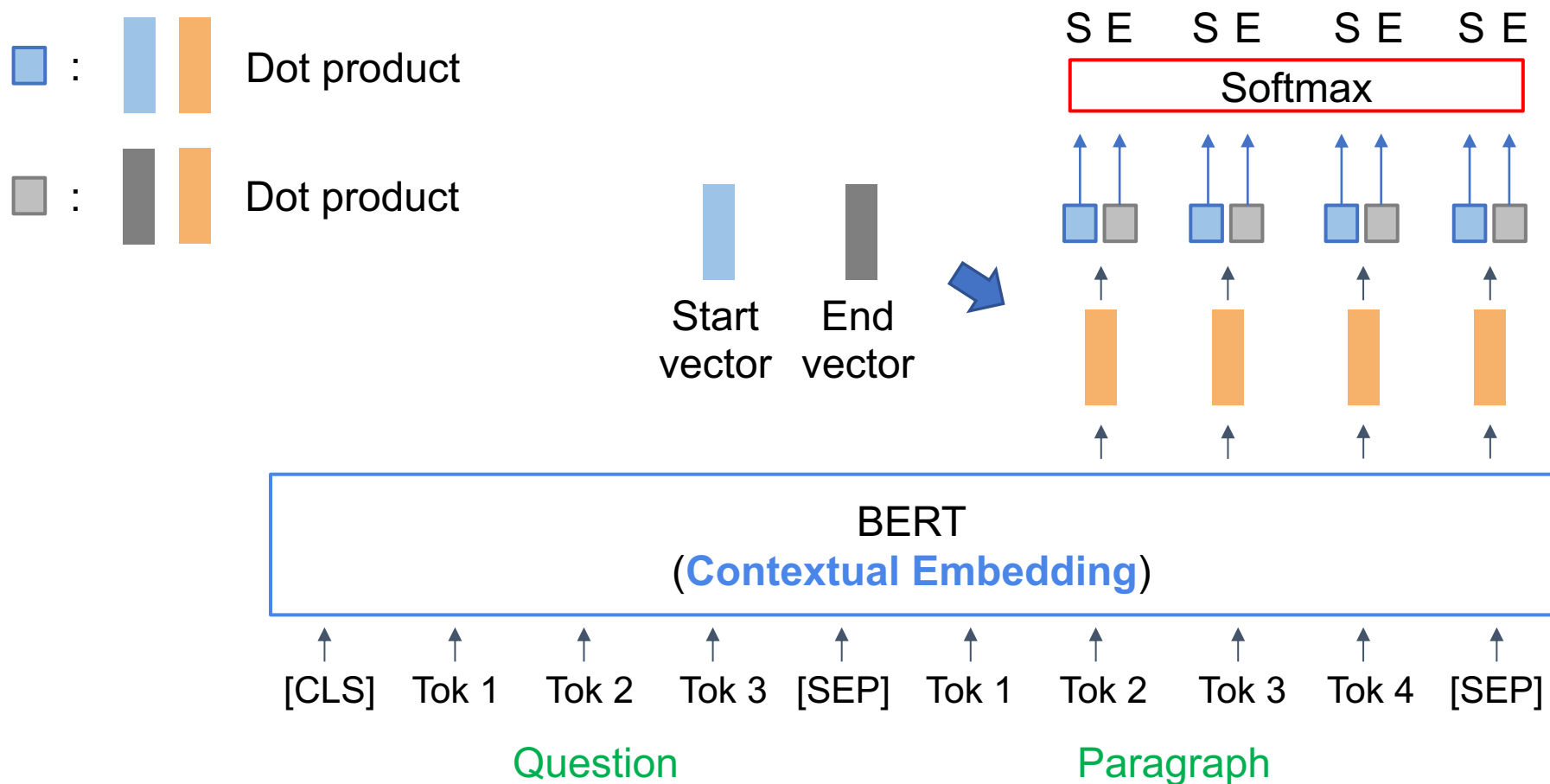
任務：Question Answering / 閱讀理解

Question: To whom did the Virgin Mary allegedly appear in 1858 in Lourdes France?

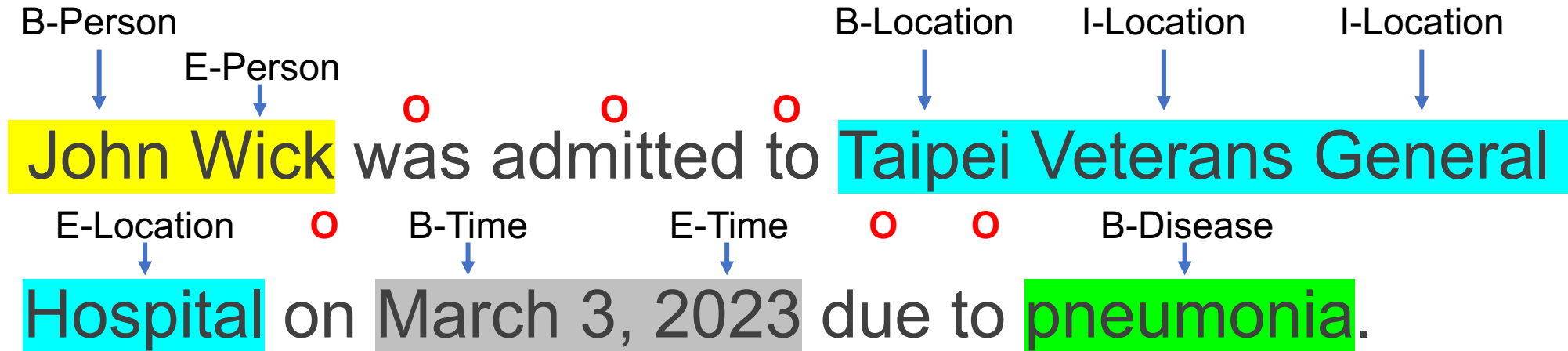
Context: Architecturally, the school has a Catholic character. Atop the Main Building's gold dome is a golden statue of the Virgin Mary. Immediately in front of the Main Building and facing it, is a copper statue of Christ with arms upraised with the legend "Venite Ad Me Omnes". Next to the Main Building is the Basilica of the Sacred Heart. Immediately behind the basilica is the Grotto, a Marian place of prayer and reflection. It is a replica of the grotto at Lourdes, France where the Virgin Mary reputedly appeared to **Saint Bernadette Soubirous** in 1858. At the end of the main drive (and in a direct line that connects through 3 statues and the Gold Dome), is a simple, modern stone statue of Mary.

Answer: { "text": ["Saint Bernadette Soubirous"], "answer_start": [515] }

Fine-tuning BERT (Question Answering)

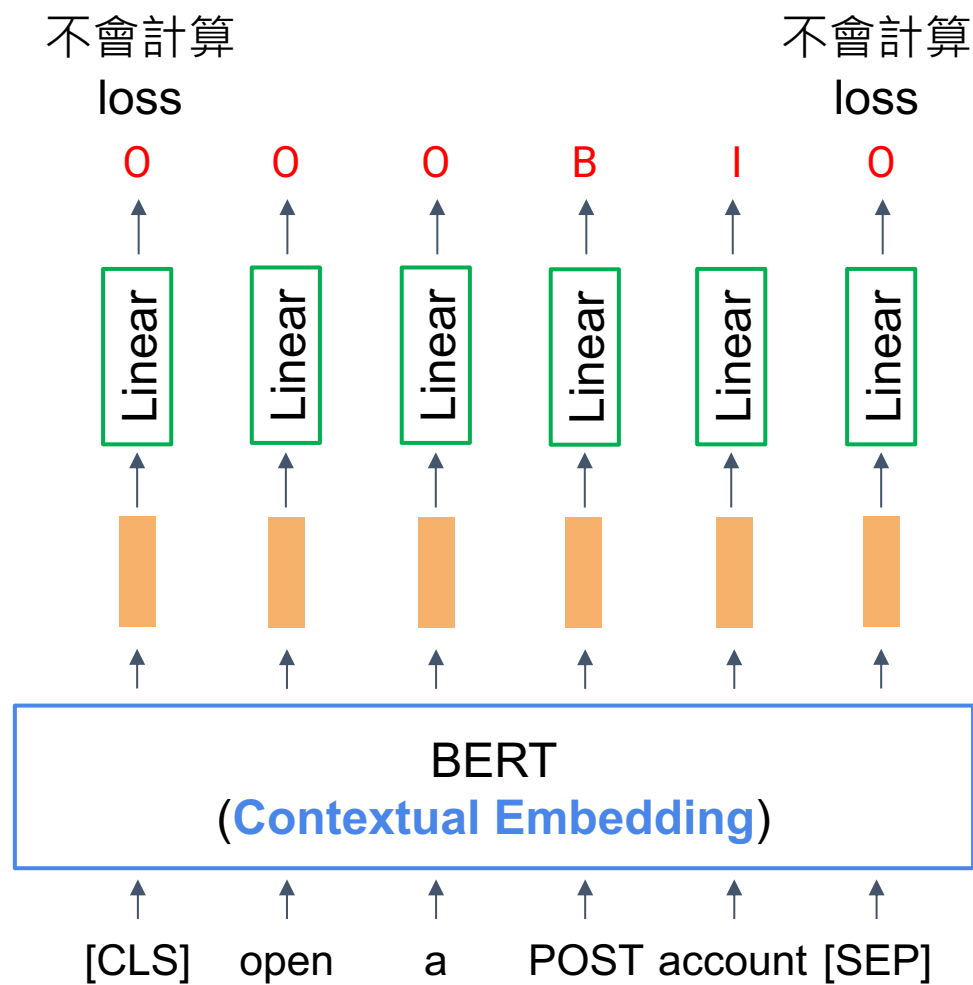


任務：命名實體辨識 (Named Entity Recognition, NER)



	Meaning
B	Beginning
I	Inside
O	Outside
E	End

Fine-tuning BERT (NER)



BERT and its Family

- ELMo
- BERT
- RoBERTa
- SpanBERT
- DeBERTa

RoBERTa: A Robustly Optimized BERT Pretraining Approach

- 和 BERT 的結構完全一樣
- 預訓練移除 Next Sentence Prediction (NSP) , 且使用更多的資料進行預訓練
- Dynamic masking (BERT 全部的資料集只做一次隨機 masking)
- 更好的超參數 (batch size, learning_rate, num_epochs)
- 實務上確實效能通常比 BERT 還要好

RoBERTa was rejected by ICLR.

Citations: 26136
(2026/03/27)

[← Go to ICLR 2020 Conference homepage](#)

RoBERTa: A Robustly Optimized BERT Pretraining Approach

Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, Veselin Stoyanov

26 Sept 2019 (modified: 12 Oct 2025) ICLR 2020 Conference Blind Submission Readers:  Everyone [Show Bibtex](#) [Show Revisions](#)

Original Pdf: [↓ pdf](#)

Community Implementations: [\[x\] 31 code implementations](#)

Keywords: Deep learning, language representation learning, natural language understanding

TL;DR: We evaluate a number of design decisions when pretraining BERT models and propose an improved recipe that achieves state-of-the-art results on many natural language understanding tasks.

Abstract: Language model pretraining has led to significant performance gains but careful comparison between different approaches is challenging. Training is computationally expensive, often done on private datasets of different sizes, and, as we will show, hyperparameter choices have significant impact on the final results. We present a replication study of BERT pretraining (Devlin et al., 2019) that carefully measures the impact of many key hyperparameters and training data size. We find that BERT was significantly undertrained, and can match or exceed the performance of every model published after it. Our best model achieves state-of-the-art results on GLUE, RACE, SQuAD, SuperGLUE and XNLI. These results highlight the importance of previously overlooked design choices, and raise questions about the source of recently reported improvements. We release our models and code.

Add [Public Comment](#)

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8 Replies

[\[-\]](#) Paper Decision

ICLR 2020 Conference Program Chairs

20 Dec 2019, 07:38 (modified: 20 Dec 2019, 08:15) ICLR 2020 Conference Paper855 Decision Readers:  Everyone [Show Revisions](#)

Decision: Reject

Comment: This paper conducts an extensive study of training BERT and shows that its performance can be improved significantly by choosing a better training setup (e.g., hyperparameters, objective functions). I think this paper clearly offers a better understanding of the importance of tuning a language model to get the best performance on downstream tasks. However, most of the findings are obvious (careful tuning helps, more data helps). I think the novelty and technical contributions are rather limited for a conference such as ICLR. These concerns are also shared by all the reviewers. The review scores are borderline, so I recommend to reject the paper.

Add [Public Comment](#)

BERT and its Family

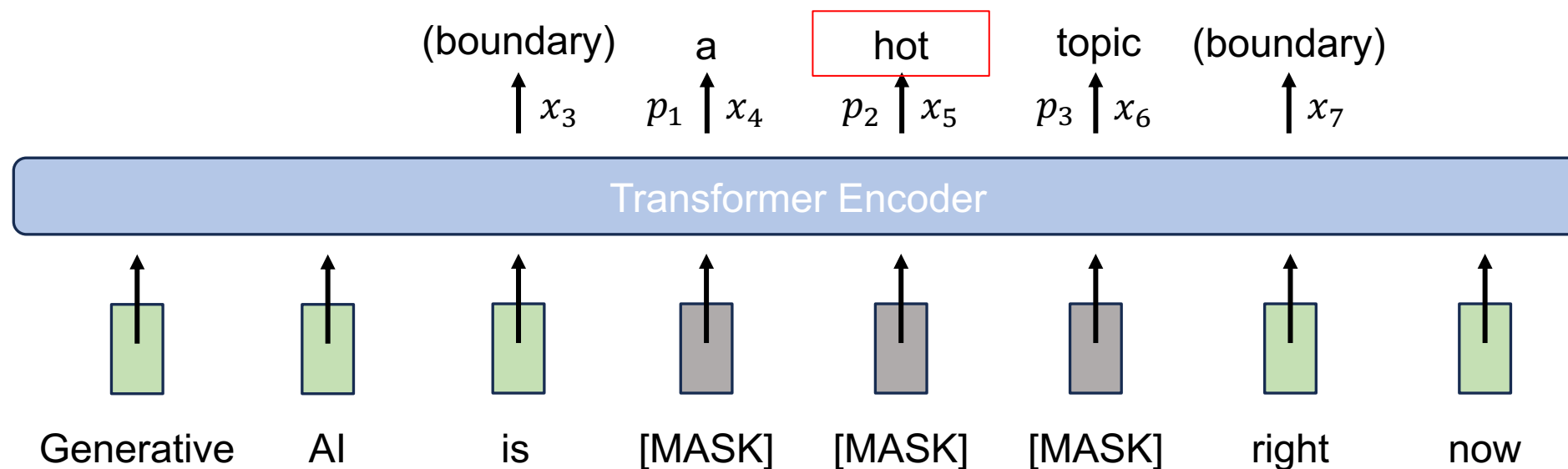
- ELMo
- BERT
- RoBERTa
- SpanBERT
- DeBERTa

SpanBERT

<https://arxiv.org/pdf/1907.10529>

基於MLM的新目標函數: Span Boundary Objective (SBO)

➡ 一次 mask 一段 span，每次的目標為還原 span 中的其中一個字



$$\mathcal{L}(\text{hot}) = \mathcal{L}_{MLM}(\text{hot}) + \mathcal{L}_{SBO}(\text{hot}) = -\log P(\text{hot} | x_5) - \log P(\text{hot} | x_3, x_7, p_2)$$

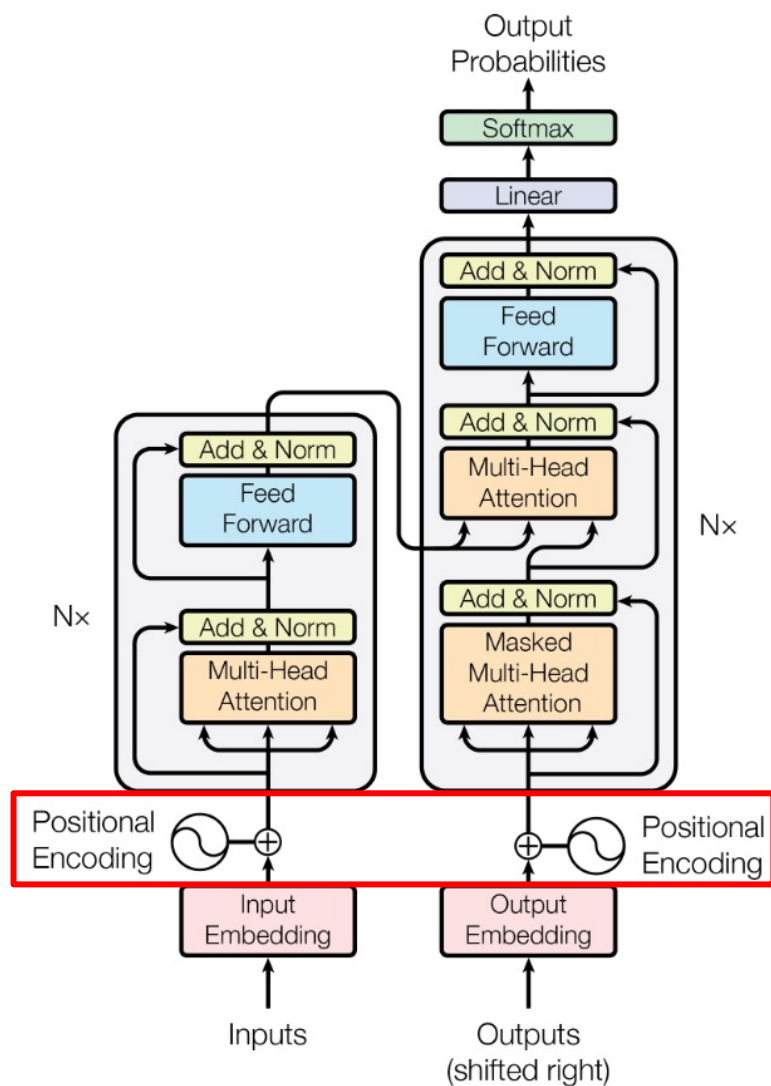
Extensions of BERT (Summary)

Model	MLM task	NSP task	Release
BERT	Static masking	Sentence relationship	2018/10
RoBERTa	Dynamic masking	Remove NSP task	2019/07
SpanBERT	Span masking	Remove NSP task	2019/07

BERT and its Family

- ELMo
- BERT
- RoBERTa
- SpanBERT
- DeBERTa

Positional Encodings 常常是加在 Input



Input	[CLS]	my	dog	is	cute	[SEP]	he	likes	play	##ing	[SEP]
Token Embeddings	$E_{[CLS]}$	E_{my}	E_{dog}	E_{is}	E_{cute}	$E_{[SEP]}$	E_{he}	E_{likes}	E_{play}	$E_{\#ing}$	$E_{[SEP]}$
	+	+	+	+	+	+	+	+	+	+	+
Segment Embeddings	E_A	E_A	E_A	E_A	E_A	E_A	E_B	E_B	E_B	E_B	E_B
	+	+	+	+	+	+	+	+	+	+	+
Position Embeddings	E_0	E_1	E_2	E_3	E_4	E_5	E_6	E_7	E_8	E_9	E_{10}

相對位置的重要性

He, Pengcheng, et al. "DeBERTa: Decoding-enhanced BERT with Disentangled Attention." ICLR 2021.

[句子1]

I am studying **deep learning** today.

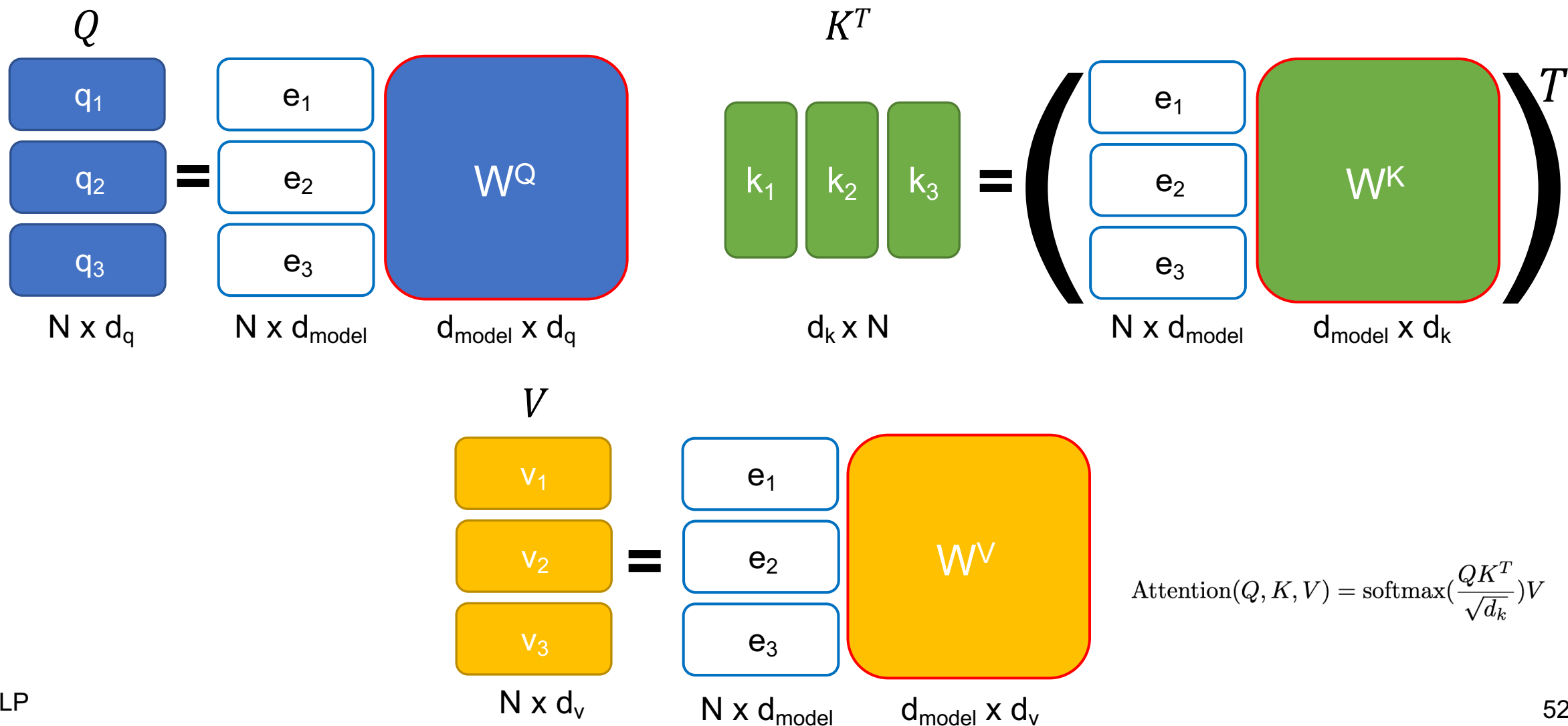
[句子2]

We discussed a **deep** neural architecture for robots, and later talked about **learning** strategies.

💡 **attention weight** 不能只由內容決定，還要用相對位置決定

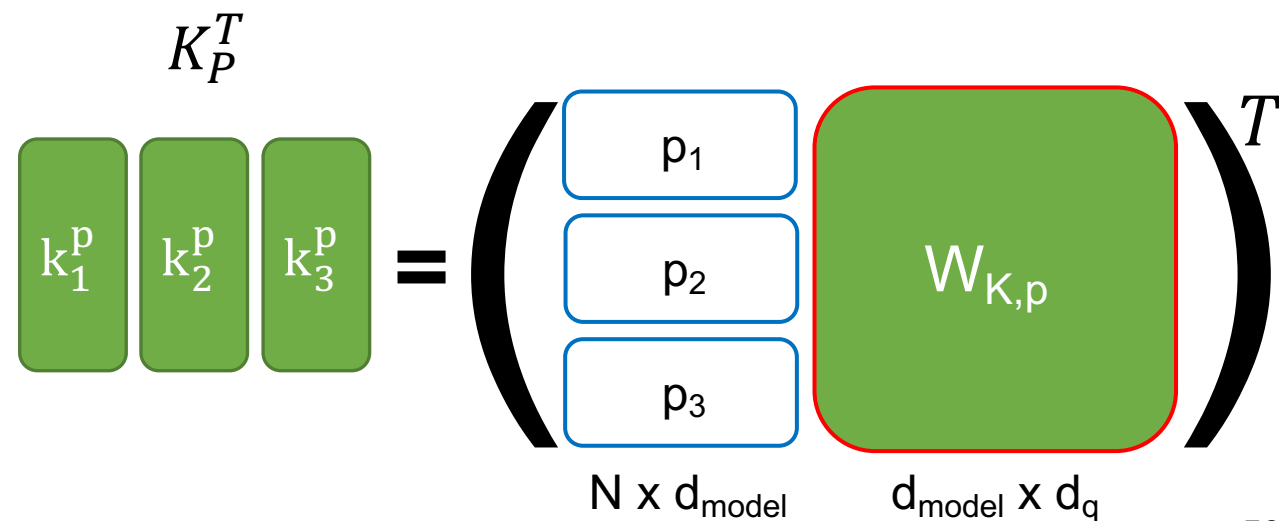
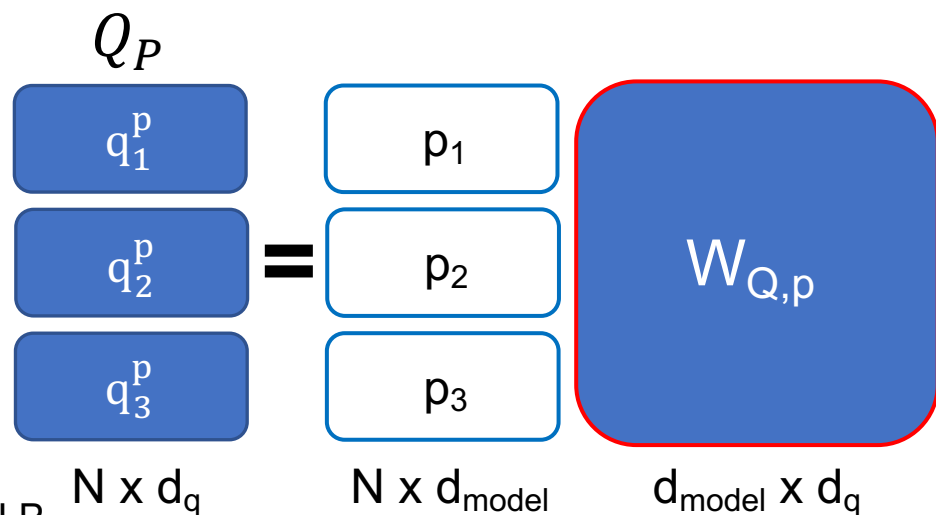
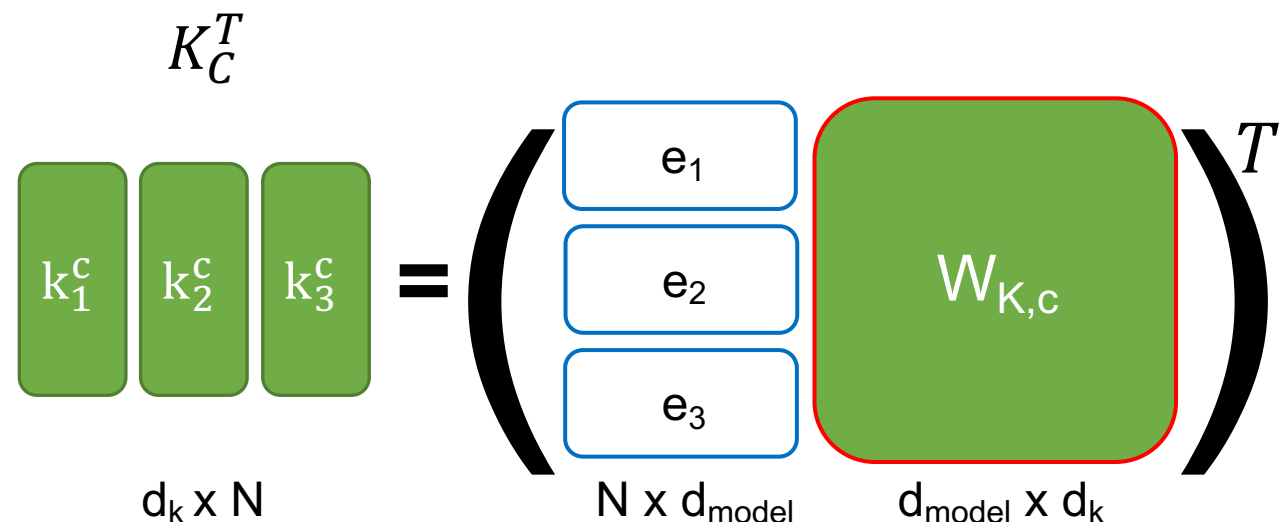
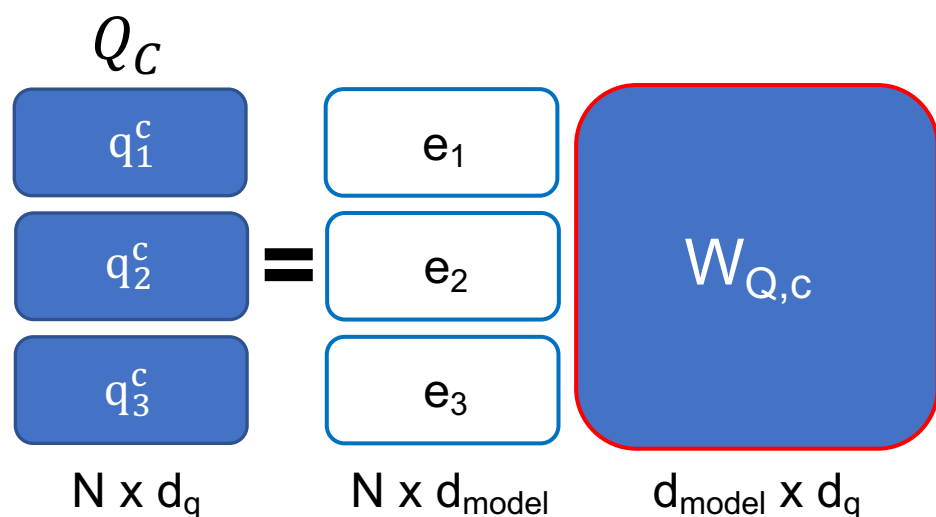
[Recap] Transformer 的權重值

可以被訓練的參數



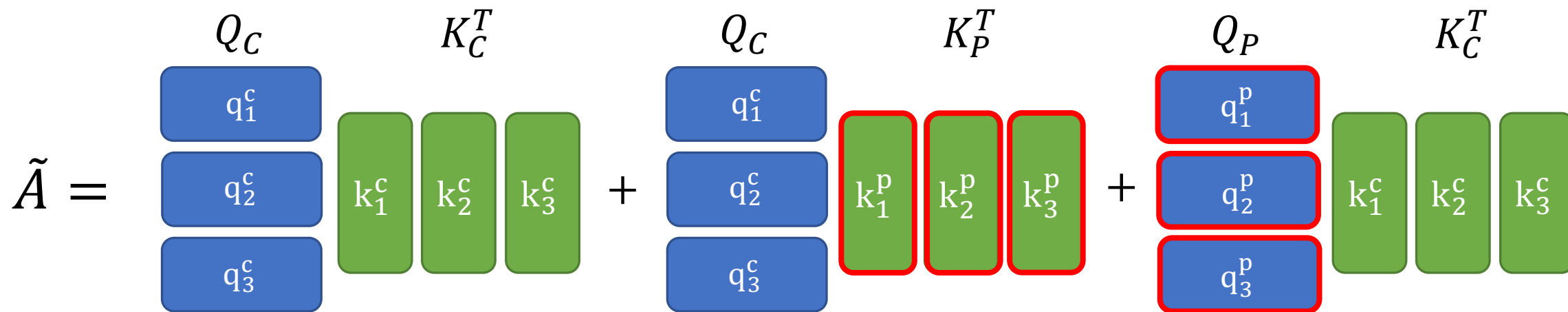
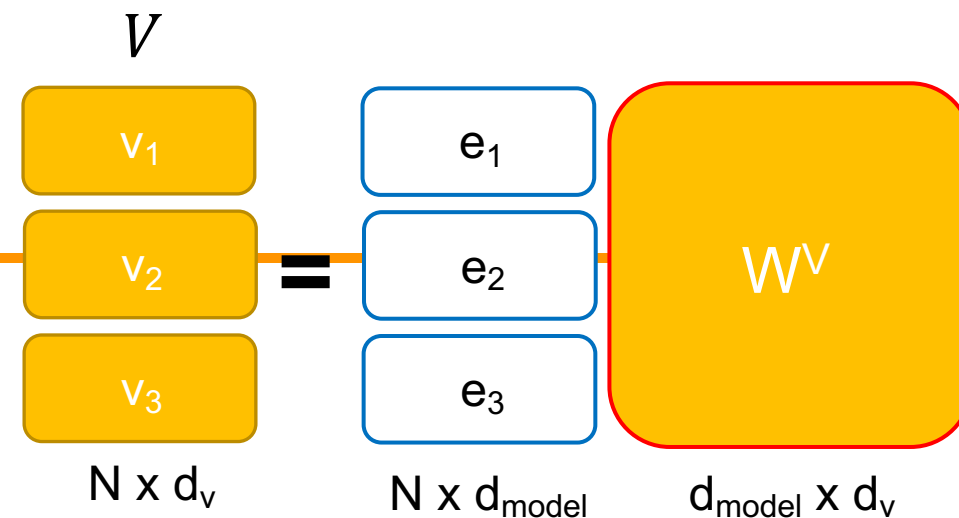
DeBERTa 的 Queries, Keys

He, Pengcheng, et al. "DeBERTa: Decoding-enhanced BERT with Disentangled Attention." ICLR 2021.



DeBERTa Attention

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$



(本頁的 positional embedding 對象有經過簡化)

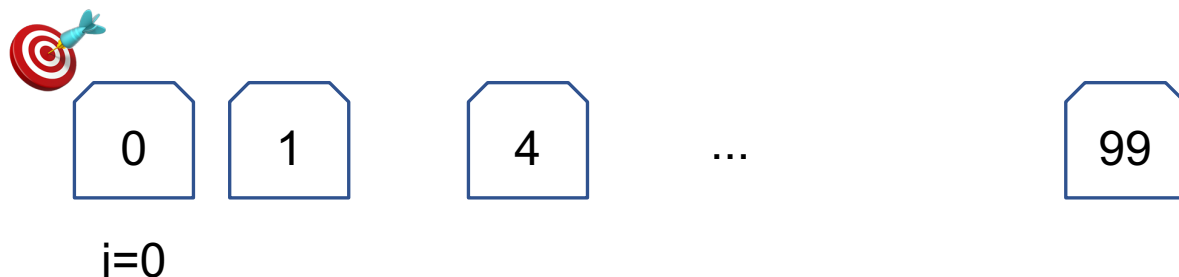
$$\text{DeBERTa_attn} = \text{softmax}\left(\frac{\tilde{A}}{\sqrt{3d}}\right)V_c$$

Positional Embedding 要取哪一個？

總 PE 的形狀為 (num_positions, embed_size)

假設 $k=4$

$$\delta(i, j) = \begin{cases} 0 & \text{for } i - j \leq -k \\ 2k - 1 & \text{for } i - j \geq k \\ i - j + k & \text{others.} \end{cases}$$



$j=0$ 時，取 index 為 4 的 PE

$j=1$ 時，取 index 為 3 的 PE

$j=4$ 時，取 index 為 0 的 PE

$j=99$ 時，也取 index 為 0 的 PE

Why there's no $Q_P \times K_P^T$?

💡 「設定上」是相對距離和語義內容具備關聯性

➡ 相對距離和相對距離理論上不應貢獻資訊量

Thank you!

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